

Eventual unimodality of independence polynomials of finite forests

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Abstract

For a finite forest F , let

$$I_F(x) = \sum_{k=0}^{\alpha(F)} i_k(F)x^k$$

be its independence polynomial, where $i_k(F)$ counts the independent k -subsets of $V(F)$ and $\alpha(F)$ is the independence number. We prove that there is an integer N_0 such that the coefficient sequence of I_F is unimodal whenever $|V(F)| \geq N_0$. Given a hypothetical counterexample, we consider the first index at which its coefficient sequence rises after an earlier fall and tilt the hard-core measure so that its mean is this index. We show that its activity is less than 27 and that its variance diverges with the order. After rooting the components, a martingale variance decomposition yields a vertex contributing a fixed fraction of the total variance. At this vertex, the difference between the conditional means under occupation and vacancy has a linear lower bound in the forest order, while the conditional occupation probability stays bounded away from zero. An exact alternating path expansion, on the other hand, shows that this difference is sublinear in the order of the descendant subtree, uniformly over all finite rooted trees at bounded activity. This contradiction proves the theorem. The argument treats the forest at one common activity rather than through componentwise products; any remaining counterexample must lie below the resulting threshold.

Keywords. Independence polynomial, unimodality, finite forest, hard-core model, martingale central limit theorem, local limit theorem.

AI disclosure. This work was developed with extensive assistance from large language model systems. Please see the Statement on AI-assisted research before the appendices.

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Introduction

The independence polynomial, introduced by Gutman and Harary [GH83], records the number of independent vertex sets of each cardinality; see [LM05] for a survey of its basic properties. Evaluated at a positive activity z , it is also the partition function of the hard-core lattice gas; the corresponding probability law assigns weight proportional to $z^{|S|}$ to each independent set S [SS05]. This probabilistic interpretation is essential below, since coefficient comparisons will be converted into local curvature statements for the cardinality of a random hard-core independent set.

Independent-set sequences need not be unimodal for general graphs. Alavi, Malde, Schwenk, and Erdős asked whether the independence polynomial of every tree, or perhaps every forest, is unimodal [AMSE87]; this question is also known as Erdős Problem 993. We establish the following asymptotic statement for forests.

Main Theorem. *There exists an integer N_0 such that the coefficient sequence of the independence polynomial of every finite forest with at least N_0 vertices is unimodal.*

In particular, the same conclusion holds for every connected tree of order at least N_0 . The proof treats a hypothetical nonunimodal forest as a whole, at one first-recovery index and one common hard-core activity. Consequently, every counterexample to Erdős Problem 993 has bounded order. This reduction is qualitative: the proof does not make the order bound computationally explicit, and finitely many forest orders remain untreated.

Several substantial subclasses were known before the present theorem. Hamidoune proved log-concavity of the independent-set sequence for claw-free graphs [Ham90], and Chudnovsky and Seymour later proved the stronger real-rootedness of their independence polynomials [CS07]. For trees this covers paths, since a tree is claw-free exactly when it has maximum degree at most two. Levit and Mandrescu proved unimodality for several classes of well-covered trees [LM03], while certain nonregular caterpillars satisfying explicit pendant-multiplicity hypotheses, specified leaf-attachment families, and other recursively described classes were treated in [BES18, GH18, ZC19]. More recently, Li, Li, Yang, and Zhang proved log-concavity for all spiders [LLYZ25]. These results use family-specific coefficient arguments, symmetric-function methods, or controlled branching geometry and do not give a uniform argument for arbitrary trees.

There are also partial results that locate where a counterexample could occur. Levit and Mandrescu proved that the last third of the independent-set sequence of every bipartite graph, hence every tree, is decreasing [LM06]. The bipartite double count behind their result is also the starting point of the activity localization in Proposition 2.2. Basit and Galvin proved that a uniformly random labelled tree has long initial increasing and terminal decreasing segments with probability tending to one [BG21], and Heilman obtained a related exponentially high-probability initial segment [Hei25]. These random-tree results leave an intermediate window and do not apply uniformly to all trees. Exact computation established log-concavity through order twenty [YMK21].

The tempting strengthening from unimodality to log-concavity is false. Kadrawi and Levit found examples beginning at order twenty-six and constructed two infinite non-log-concave tree families [KL23]. Galvin found failures much farther into the coefficient sequence [Gal26], and Ramos and Sun used an AI-guided search to produce a large experimental census of further examples [RS25]. Linear-recurrence constructions now give families with

several consecutive log-concavity failures [BGG26]. Nevertheless, the original Kadrawi–Levit families are unimodal [Li26], and symmetric unimodal tree polynomials have also been constructed [HKV26]. These developments show that log-concavity is not the missing invariant: substantial local failures of log-concavity can coexist with unimodality.

The probabilistic language of the proof belongs to the hard-core-model tradition. The occupancy-ratio recursion on a rooted tree and attenuation of boundary influence are classical; see Weitz [Wei06] and the survey of Davies and Kang [DK25]. The alternating path identity used in Section 5 is obtained by logarithmically differentiating this recursion. The martingale central limit theorem used in the variance argument is the constant-variance form of Hall and Heyde [HH80]. Jain, Perkins, Sah, and Sawhney developed hard-core local central limit estimates by Fourier inversion below the bounded-degree uniqueness threshold [JPSS22]. Recent work studies bounds on the derivative of the occupancy fraction with respect to the logarithm of the activity [DST25, ZX26]. These bounded-degree results provide the closest analytic precedents for the present argument. Here the degrees may be unbounded, the activity is selected from the first recovery of the original forest, and the local-limit analysis must determine the sign of a three-term log-concavity determinant rather than only a point probability.

Our argument applies uniformly across all finite forest topologies and proceeds directly through the hard-core measure rather than through real-rootedness or log-concavity. Suppose that (F_n, s_n, z_n, V_n) is a sequence of canonical first-recovery states with $|V(F_n)| \rightarrow \infty$. Section 2 shows that $0 < z_n < 27$ and $V_n \rightarrow \infty$. After rooting every component, the centered occupation count has the martingale representation

$$X_n - \mathbb{E}X_n = \sum_{u \in F_n} \Delta_u \eta_u, \quad V_n = \sum_{u \in F_n} a_u p_u q_u \Delta_u^2.$$

Theorem 4.1 gives a vertex u_n for which one summand in the variance identity is at least $\kappa_{27} V_n$. Theorems 4.2 and 4.3 then imply $p_{u_n} \geq \rho$ and $|\Delta_{u_n}| \geq \delta |V(F_n)|$, for constants $\rho, \delta > 0$. On the other hand, logarithmic differentiation of the rooted-tree recursion gives

$$\Delta_{u_n} = \sum_{x \in T_{u_n}} (-1)^{d(u_n, x)} \prod_{y \in P(u_n, x) \setminus \{u_n\}} p_y,$$

where $P(u_n, x)$ is the unique path from u_n to x . Since $z_n < 27$, all factors in the product are uniformly bounded away from one. The lower bound on p_{u_n} bounds the number of descendants for which the path product exceeds any fixed threshold. Lemma 5.1 consequently gives $|\Delta_{u_n}| = o(|T_{u_n}|)$, uniformly over the relevant rooted trees. Since $|T_{u_n}| \leq |V(F_n)|$, this contradicts the preceding linear lower bound. The argument is qualitative: it gives no effective value of N_0 .

The analytic bridge used in the proof is stronger than weak convergence. If integer-valued random variables have integer means s_n and variances $V_n \rightarrow \infty$, and if their normalized characteristic functions, extended by zero outside their expanding lattice fundamental domains, converge in quadratically weighted $L^1(\mathbb{R})$ to the characteristic function of a twice continuously differentiable density f , then Fourier inversion gives

$$V_n^2 [\Pr(X_n = s_n)^2 - \Pr(X_n = s_n - 1) \Pr(X_n = s_n + 1)] \rightarrow (f'(0))^2 - f(0)f''(0).$$

The exact two-variable inversion identity is proved in (B.16)–(B.21). This weighted full-domain convergence, rather than a central limit theorem alone, is what transfers a limiting density to adjacent coefficient curvature.

Appendices A and B establish the uniform Fourier and local-curvature estimates. Appendices C and D develop the first-recovery bounds and the fixed variance fraction theorem used in the main argument. Structurally, the proof rests on the Doob variance decomposition for a parent-before-child reveal order and on uniform attenuation in the rooted occupation recursion.

1. Independence polynomials and first recovery

All graphs in the paper are finite and simple. A forest need not be connected, and an isolated vertex is allowed. Let F be a finite forest of order N , write $i_k = i_k(F)$, and put $\alpha = \alpha(F)$. We set $i_k = 0$ outside $0 \leq k \leq \alpha$. A finite sequence is **unimodal** if there is an index m such that it is nondecreasing through m and nonincreasing thereafter. Equivalently, it never strictly rises after it has strictly fallen. If the coefficient sequence of I_F is not unimodal, define its **first-recovery index** by

$$s = s_*(F) := \min\{k \in \{0, \dots, \alpha - 1\} : i_k < i_{k+1} \text{ and } i_j > i_{j+1} \text{ for some } 0 \leq j < k\}.$$

Minimality gives

$$i_{s-1} \geq i_s < i_{s+1}, \quad i_s^2 < i_{s-1}i_{s+1}. \quad (1.1)$$

For $z > 0$, define the tilted hard-core law

$$\Pr_z(X = k) = \frac{i_k z^k}{I_F(z)},$$

and write

$$K_1(I_F; z) = \mathbb{E}_z X = \frac{z I_F'(z)}{I_F(z)}, \quad K_2(I_F; z) = \text{Var}_z X.$$

Since

$$\frac{d}{d \log z} K_1(I_F; z) = K_2(I_F; z) > 0,$$

the function $K_1(I_F; z)$ is strictly increasing. Its limits as $z \downarrow 0$ and $z \rightarrow \infty$ are 0 and $\alpha(F)$, respectively, and (1.1) gives $1 \leq s \leq \alpha(F) - 1$. Hence there is a unique activity $z_s > 0$ for which $K_1(I_F; z_s) = s$. The quadruple (F, s, z_s, V) , where $V = K_2(I_F; z_s)$, is called the **canonical first-recovery state** of F . A **canonical first-recovery sequence** is a sequence (F_n, s_n, z_n, V_n) of such states, one for each member of a sequence of nonunimodal finite forests. We call the sequence **variance-divergent** if $V_n \rightarrow \infty$.

At this activity, the second inequality in (1.1) is exactly the strict reverse log-concavity inequality for the three central tilted probabilities:

$$\Pr_z(X = s)^2 < \Pr_z(X = s - 1) \Pr_z(X = s + 1). \quad (1.2)$$

Throughout, first recovery refers only to the original forest at its common activity; no component or boundary-conditioned descendant subtree is assigned a recovery property.

2. Localization of a canonical counterexample

We first prove the quantitative reductions used at the end of the argument.

Proposition 2.1 (index and variance-size localization). For every nonunimodal N -vertex forest and its first-recovery state,

$$s > \frac{\sqrt{N}}{2}, \quad V \geq \frac{\sqrt{N}}{8(1+Z)^4} \quad (2.1)$$

whenever $z_s \leq Z$.

Proof. For the index bound, if $1 \leq k \leq \lfloor \sqrt{N}/2 \rfloor$, every nonindependent k -set contains an edge. Since a forest has at most $N - 1$ edges, the union bound gives

$$i_k \geq \binom{N}{k} - (N - 1) \binom{N - 2}{k - 2} = \binom{N}{k} \left(1 - \frac{k(k - 1)}{N} \right) > \frac{3}{4} \binom{N}{k},$$

whereas, whenever this integer range is nonempty (and hence $N \geq 4$),

$$i_{k-1} \leq \binom{N}{k-1} = \binom{N}{k} \frac{k}{N - k + 1} \leq \frac{1}{3} \binom{N}{k}.$$

Thus $i_{k-1} < i_k$ throughout that integer range. The first strict fall must therefore occur at an index $j \geq \lfloor \sqrt{N}/2 \rfloor$, and its later recovery satisfies $s > j$. Since s is an integer, this proves $s > \sqrt{N}/2$.

For the variance bound, let H be the set of vertices of degree at least three and let L be the set of vertices of degree one. Summing the identity $\sum_v (\deg v - 2) = -2$ over the nontrivial components shows that

$$\sum_{v \in H} (\deg v - 2) = |L| - 2q,$$

where q is the number of nontrivial components. Hence $|H| \leq |L| \leq N - |H|$, so $|H| \leq N/2$. At least $N/2$ vertices therefore have degree at most two. One color class of a bipartition of the forest contains an independent set S of at least $N/4$ such vertices.

Condition on the occupation variables outside S . The variables in S are then conditionally independent. A vertex in S has conditional variance $z/(1+z)^2$ when all its neighbors are absent, and zero otherwise. For any vertex v ,

$$I_G(z) = I_{G-v}(z) + z I_{G-N[v]}(z) \leq (1+z) I_{G-v}(z),$$

so $\Pr_z(v \notin I) \geq (1+z)^{-1}$. More explicitly, if $U = \{u_1, \dots, u_m\}$ with $m \leq 2$, apply the displayed inequality first to G , then to $G - u_1$, and so on. This gives $I_G(z) \leq (1+z)^m I_{G-U}(z)$,

and hence $\Pr_z(U \cap I = \emptyset) \geq (1+z)^{-m}$. Apply this with $U = N(v)$, whose size is at most two. The conditional-variance identity now gives

$$K_2(I_F; z) \geq \frac{Nz}{4(1+z)^4}. \quad (2.2)$$

At the canonical activity, the elementary occupation bound $K_1(I_F; z) \leq Nz/(1+z)$ and the index bound give $z > 1/(2\sqrt{N})$. Substituting this and $z \leq Z$ into (2.2) proves (2.1). $\square$

Proposition 2.2 (uniform activity localization). Every first-recovery state of a finite forest satisfies

$$z_s < 27. \quad (2.3)$$

Proof. Following the bipartite double-counting argument behind [LM06], let α be the independence number, and fix a bipartition with color classes C_1, C_2 . For an independent k -set J , let $A_r(J)$ be the vertices in $C_r \setminus J$ that can be added to J . The set $J \cup A_r(J)$ is independent, because C_r is independent and every vertex of $A_r(J)$ has no neighbor in J . Consequently $|A_r(J)| \leq \alpha - k$ for $r = 1, 2$. Counting pairs (J, v) for which J is an independent k -set and $J \cup \{v\}$ is independent gives

$$(k+1)i_{k+1} = \sum_{J: |J|=k} (|A_1(J)| + |A_2(J)|) \leq 2(\alpha - k)i_k.$$

Thus a strict rise at k implies

$$k \leq b := \left\lfloor \frac{2\alpha - 2}{3} \right\rfloor,$$

so $s \leq b$. If $\nu = N - \alpha$ is the matching number, König's theorem [LP86] and the three choices on each matching edge give

$$I_F(1) \leq 3^\nu 2^{\alpha-\nu} \leq 3^\alpha.$$

Put $d = \alpha - b$. At $z = 27$,

$$I_F(27)(K_1(I_F; 27) - b) = \sum_j (j - b)i_j 27^j \geq 27^b (d27^d - b3^\alpha) > 0.$$

For $\alpha = 3r, 3r+1, 3r+2$, the quotient $d27^d/3^\alpha$ is respectively $27(r+1)$, $9(r+1)$, and $81(r+2)$, each larger than b . Hence $K_1(I_F; 27) > b \geq s$, and strict monotonicity of the tilted mean proves (2.3). $\square$

Combining Propositions 2.1 and 2.2, any sequence of nonunimodal forests with orders tending to infinity produces canonical states with

$$0 < z_n < 27, \quad V_n \geq \frac{\sqrt{N_n}}{8 \cdot 28^4} \longrightarrow \infty. \quad (2.4)$$

3. Rooted-tree recursion and the variance decomposition

Let I be the random independent set under the global hard-core law. Fix one root in each component of F , orient every edge away from its component root, and let $\mathcal{R}(F)$ be the set of component roots. Let T_x denote the subtree induced by x and all of its descendants. Define

$$P_x(t) = I_{T_x}(t), \quad Q_x(t) = I_{T_x-x}(t), \quad A_x(t) = t \prod_{y \text{ child of } x} Q_y(t).$$

Here P_x is the partition function with the parent of x absent, Q_x is the partition function with x absent, and A_x is the occupied contribution. The rooted-tree recursion is

$$Q_x(t) = \prod_{y \text{ child of } x} P_y(t), \quad P_x(t) = Q_x(t) + A_x(t).$$

Define the occupation odds, conditional occupation probability, and vacancy probability by

$$R_x = \frac{A_x(z)}{Q_x(z)}, \quad p_x = \frac{R_x}{1 + R_x}, \quad q_x = 1 - p_x.$$

Equivalently, $p_x = \Pr_z(x \in I \mid \text{par}(x) \notin I)$, with the virtual absent-parent convention at a component root, and $R_x = z \prod_{y \text{ child of } x} q_y$.

Define the **conditional mean difference** at x by

$$\Delta_x = \mathbb{E}_z(|I \cap T_x| \mid x \in I) - \mathbb{E}_z(|I \cap T_x| \mid x \notin I).$$

If a_x denotes the probability that the parent of x is absent in the global hard-core law, with a virtual absent parent and $a_r = 1$ for every $r \in \mathcal{R}(F)$, define the **vertex variance contribution** at x by

$$g(x) = a_x p_x q_x \Delta_x^2. \tag{3.1}$$

Write $\xi_x = \mathbf{1}_{\{x \in I\}}$, with $\xi_{\text{par}(r)} = 0$ for a component root, and set

$$\eta_x = \xi_x - p_x(1 - \xi_{\text{par}(x)}).$$

Since p_x is this conditional occupation probability, the variables η_x are martingale differences in a parent-before-child ordering, with $\mathbb{E}\eta_x^2 = a_x p_x q_x$. The recursion $\Delta_x + \sum_{y \text{ child of } x} p_y \Delta_y = 1$ makes the coefficient of each ξ_x in $\sum_x \Delta_x \eta_x$ equal to one. Thus $\sum_x \Delta_x \eta_x = X - \sum_x p_x \Delta_x$; taking expectations gives $\mathbb{E}X = \sum_x p_x \Delta_x$, and hence

$$X - \mathbb{E}X = \sum_{x \in F} \Delta_x \eta_x.$$

Thus each vertex contributes the orthogonal variance $g(x)$, and

$$V = \sum_{x \in F} g(x). \tag{3.2}$$

The boundary-conditioned hard-core laws on T_x , taken with this fixed rooting, determine p_x, q_x, Δ_x , while a_x records the position of T_x in its rooted component.

4. A vertex carrying a fixed variance fraction and the scale of first recovery

Theorem 4.1 (a vertex carrying a fixed variance fraction). There are constants $\kappa_{27} > 0$ and $V_{27} < \infty$ such that every canonical first-recovery state with $0 < z < 27$ and $V \geq V_{27}$, under every choice of one root in each forest component, contains a vertex u satisfying

$$g(u) \geq \kappa_{27}V. \quad (4.1)$$

Theorem 4.1 is Theorem D.1. Its key reduction is a finite-branch approximation along an arbitrary variance-divergent canonical sequence. Lemma D.4 rules out an exposed antichain of individually negligible descendant variance masses carrying a fixed fraction of the total variance: such an antichain would produce a Gaussian-mixture submeasure with positive density at the mean, whereas Lemmas C.4 and C.5 give $\sqrt{V} \Pr_z(X = s) \rightarrow 0$. Lemma D.5 therefore captures, for every fixed $\varepsilon > 0$, all but εV of the variance on an ancestor-closed rooted subforest with a bounded number of root-to-leaf branches, retaining the coefficients of the global martingale. If every vertex contribution were $o(V)$, a diagonal choice could let ε tend to zero while keeping the branch count controlled relative to the largest vertex contribution. The large-displacement estimate would then give the martingale Lindeberg condition, and the branch-count control gives concentration of the predictable quadratic variation. The resulting Gaussian limit, together with the full-domain Fourier estimate, yields a positive adjacent log-concavity determinant at the mean, contrary to the first-recovery sign. Thus a large canonical counterexample must instead have a single macroscopic variance contribution, as asserted in (4.1).

Theorem 4.2 (occupation and displacement bounds at a dominant vertex). For every $c > 0$ there are constants $\rho(c) > 0$ and $V_{\text{bal}}(c) < \infty$ such that, in every canonical first-recovery state with $0 < z < 27$ and $V \geq V_{\text{bal}}(c)$, a vertex satisfying $g(u) \geq cV$ also satisfies

$$p_u \geq \rho(c), \quad 4cV \leq \Delta_u^2 \leq \frac{784}{\rho(c)}V. \quad (4.2)$$

The lower bound on p_u is the substantive part of Theorem C.3. If $p_u \rightarrow 0$ while $g(u) \geq cV$, then $\Delta_u^2/V \rightarrow \infty$. The recursion $\Delta_u = 1 - \sum_v p_v \Delta_v$, combined with weighted Cauchy–Schwarz, charges such a large displacement to variance in the children. Lemma C.2 formalizes this charging: if $b \rightarrow \infty$ and $V/b^2 \rightarrow 0$, the total variance contribution of vertices with $|\Delta_x| > b$ is $o(V)$. With $b = |\Delta_u|/2$, both conditions hold, yet u belongs to this set and contributes $g(u) \geq cV$, a contradiction.

Once $p_u \geq \rho(c)$ is known, the displacement bounds are immediate. The lower bound follows from $a_u p_u q_u \leq 1/4$ in (3.1). For the upper bound, $a_u, q_u \geq 1/28$, while (3.2) gives $g(u) \leq V$; thus $a_u p_u q_u \geq \rho(c)/784$.

Theorem 4.3 (size and variance at first recovery). There are constants $c_V, C_V, c_N, C_N > 0$ and $V_{\text{sc}} < \infty$ such that every canonical first-recovery state of a finite forest with $0 < z < 27$ and $V \geq V_{\text{sc}}$ satisfies

$$c_V s^2 \leq V \leq C_V s^2, \quad c_N s \leq N \leq C_N s. \quad (4.3)$$

Theorem 4.3 is Theorem C.7.

The quadratic scale in (4.3) is special to a hypothetical first-recovery state. Lemma C.5 makes the probability of the exact mean exponentially small. After one bipartition class is revealed, let M and W be the conditional mean and variance of the other class. The proof gives $\Pr(W \geq s/112) \geq \delta_*$ for a fixed $\delta_* > 0$. Taking $H = \gamma s$ with a sufficiently small fixed $\gamma > 0$, Lemma C.6 shows that $\{W \geq 4H, (M - s)^2 \leq HW\}$ has probability tending to zero. Thus $|M - s|$ is at least a fixed positive multiple of s with nonvanishing probability, and the law of total variance gives $V \gtrsim s^2$. The matching upper bound follows from $N = O(s)$ and $X^2 \leq NX$.

Lemma C.1 first places every such state of sufficiently large variance in the range $3/2 < z < 27$, by combining the low-activity Gaussian limit with the local-curvature contradiction. In this range, any componentwise rooting has at least half of its vertices with at most one child. Each such vertex has a uniformly positive global occupation probability, which gives $N = O(s)$; the reverse inequality $s \leq N$ is immediate.

Apply Theorem 4.2 with $c = \kappa_{27}$ and set $\rho = \rho(\kappa_{27})$. The immediate consequence of (4.1)–(4.3) is the fixed linear lower bound

$$|\Delta_u| \geq \frac{2\sqrt{\kappa_{27}c_V}}{C_N} N =: \delta N. \quad (4.4)$$

5. A sublinear bound for the conditional mean difference

The remaining input is local and uses only the rooted recursion. Taking absolute values in the alternating path expansion reduces the problem to controlling the total path weight. Bounded activity makes every path product decay geometrically with depth. At the same time, if the conditional occupation probability at a vertex is bounded below, the recursion permits only boundedly many children whose occupation probabilities exceed a fixed threshold. Consequently, for each threshold only boundedly many descendants can carry a nonnegligible path weight; every other descendant contributes at most that threshold. Lemma 5.1 makes this observation uniform.

Lemma 5.1 (sublinear conditional mean difference). Fix $0 < Z < \infty$ and $\rho > 0$. If $\rho > Z/(1+Z)$, the class below is empty. Otherwise, for every $\varepsilon > 0$ there is a finite constant $K = K(Z, \rho, \varepsilon)$ such that, for every finite rooted tree, every $0 < z \leq Z$, and every vertex u with $p_u \geq \rho$,

$$|\Delta_u| \leq \varepsilon |T_u| + K. \quad (5.1)$$

Consequently, uniformly over this class,

$$\frac{|\Delta_u|}{|T_u|} \rightarrow 0 \quad \text{as } |T_u| \rightarrow \infty. \quad (5.2)$$

Proof. The rooted-tree recursion gives

$$R_x = z \prod_{y \text{ child of } x} q_y = z \prod_y (1 - p_y), \quad p_x = \frac{R_x}{1 + R_x}. \quad (5.3)$$

Since $R_x \leq z \leq Z$,

$$0 < p_x \leq \bar{p}_Z := \frac{Z}{1 + Z} < 1. \quad (5.4)$$

Moreover, logarithmic differentiation of (5.3) gives the recursion

$$\Delta_x = z \frac{d}{dz} \log R_x = 1 - \sum_{y \text{ child of } x} p_y \Delta_y. \quad (5.5)$$

Here $z d \log A_x / dz$ and $z d \log Q_x / dz$ are the conditional mean occupation counts of T_x when x is occupied and absent, respectively.

For $x \in T_u$, let $P(u, x)$ be the unique path from u to x , let $d(u, x)$ be its length, and set

$$w_u = 1, \quad w_x = \prod_{x_j \in P(u, x) \setminus \{u\}} p_{x_j},$$

the product over the nonroot vertices on the path from u to x . Induction on subtree height in (5.5) gives the exact finite expansion

$$\Delta_u = \sum_{x \in T_u} (-1)^{d(u, x)} w_x.$$

Therefore

$$|\Delta_u| \leq W_u := \sum_{x \in T_u} w_x. \quad (5.6)$$

Put $\ell(t) = -\log(1 - t)$. If $p_x \geq \sigma$, then $R_x \geq \sigma/(1 - \sigma)$, and (5.3) implies

$$\sum_{y \text{ child of } x} \ell(p_y) = \log \frac{z}{R_x} \leq \log \frac{Z(1 - \sigma)}{\sigma}. \quad (5.7)$$

Fix τ with $0 < \tau < \bar{p}_Z$. Equation (5.7) shows that a vertex with $p_x \geq \sigma$ has at most

$$M(\sigma, \tau) := \left\lceil \frac{\log(Z(1 - \sigma)/\sigma)}{-\log(1 - \tau)} \right\rceil \quad (5.8)$$

children with conditional occupation probability at least τ . Whenever $p_x \geq \sigma$, (5.4) gives $\sigma \leq \bar{p}_Z$, so the numerator in (5.8) is nonnegative.

If $w_x \geq \tau$, every conditional occupation probability on the path below u is at least τ , and (5.4) gives

$$w_x \leq \bar{p}_Z^{d(u, x)}.$$

Thus $d(u, x) \leq D_\tau$, where

$$D_\tau := \max\{d \in \mathbb{Z}_{\geq 0} : \bar{p}_Z^d \geq \tau\} < \infty.$$

At the first generation use (5.8) with $\sigma = \rho$; at every later vertex satisfying $p_x \geq \tau$ use it with $\sigma = \tau$. Hence the number of descendants with $w_x \geq \tau$, including u , is at most

$$K_0 = 1 + M(\rho, \tau) \sum_{j=0}^{D_\tau-1} M(\tau, \tau)^j. \quad (5.9)$$

Every remaining descendant has weight below τ . Splitting (5.6) gives

$$|\Delta_u| \leq W_u \leq K_0 + \tau|T_u|. \quad (5.10)$$

Given $\varepsilon > 0$, choose

$$\tau = \min\{\varepsilon/2, \bar{p}_Z/2\}$$

and set $K = K_0(Z, \rho, \tau)$. These choices depend only on Z, ρ, ε , so (5.10) proves (5.1) uniformly over the stated class. Dividing by $|T_u|$ and letting $|T_u| \rightarrow \infty$ proves (5.2). $\square$

6. Exclusion of every variance-divergent canonical sequence

Assume that (F_n, s_n, z_n, V_n) is a canonical first-recovery sequence with $V_n \rightarrow \infty$. Choose one arbitrary root in every component of each forest. For each sufficiently large n , choose a vertex supplied by Theorem 4.1 and denote it by u_n . Theorems 4.2 and 4.3 give

$$p_{u_n} \geq \rho, \quad |\Delta_{u_n}| \geq \delta N_n. \quad (6.1)$$

Use Lemma 5.1 with $Z = 27$, the fixed occupation constant ρ , and error tolerance $\varepsilon = \delta/2$. Let $K = K(27, \rho, \delta/2)$ be its constant. Regard T_{u_n} as a rooted tree with the parent of u_n forced absent. By the definitions in Section 3, its local probabilities p_x and root mean difference Δ_{u_n} agree with those above. Lemma 5.1 and $|T_{u_n}| \leq N_n$ now give

$$|\Delta_{u_n}| \leq \frac{\delta}{2}|T_{u_n}| + K \leq \frac{\delta}{2}N_n + K. \quad (6.2)$$

Since $0 \leq X_n \leq N_n$, Popoviciu's inequality gives $V_n \leq N_n^2/4$. Thus $V_n \rightarrow \infty$ implies $N_n \rightarrow \infty$. For $N_n > 2K/\delta$, the right side of (6.2) is strictly less than δN_n , contradicting (6.1). Therefore:

No canonical first-recovery sequence has $V_n \rightarrow \infty$.

(6.3)

7. Eventual unimodality

Theorem 7.1. There exists $N_0 < \infty$ such that every finite forest F with $|V(F)| \geq N_0$ has a unimodal independence-polynomial coefficient sequence.

Proof. If no such N_0 existed, there would be nonunimodal forests F_n with $N_n \rightarrow \infty$. Let (F_n, s_n, z_n, V_n) be their canonical first-recovery states. By (2.4), they satisfy $0 < z_n < 27$ and $V_n \rightarrow \infty$, contradicting (6.3). $\square$

Corollary 7.2 (connected trees). There exists $N_0 < \infty$ such that every finite connected tree T with $|V(T)| \geq N_0$ has a unimodal independence-polynomial coefficient sequence.

Proof. Every finite connected tree is a finite forest, so this is the special case of Theorem 7.1 with one component. $\square$

8. Remaining finite cases

Theorem 7.1 reduces the full finite-forest problem to finitely many orders. Indeed, (6.3) shows that the variance of every canonical first-recovery counterexample is bounded by some finite constant V_0 . Proposition 2.1 then implies

$$N \leq 64 \cdot 28^8 V_0^2. \quad (8.1)$$

The constant V_0 obtained above is not explicit, so (8.1) is not currently computationally effective and the remaining orders are not verified here. Thus the full forest conjecture, and likewise its connected-tree case, is reduced to a finite problem whose order bound is presently ineffective. The conclusion concerns unimodality; log-concavity is known to fail even for trees.

Statement on AI-assisted research

During the development of this work, the author relied extensively on large language model (LLM) systems to explore proof strategies, assist with drafting and the critique of arguments, and generate Lean 4 formalizations of selected arguments. The systems used included an author-developed LLM research system, the GPT-5.6 Pro web interface, and OpenAI Codex. The author takes responsibility for the statements, proofs, citations, and final presentation in this manuscript. A separate companion document describing the AI-assisted development of the proof and its intermediate lemmas is in preparation.

Appendix A. Uniform Fourier decay for finite forests

This appendix proves the Fourier envelope used in (B.6) and (D.21), uniformly over all finite forests, all degrees and depths, and all activities in a fixed bounded interval. It also records a weighted Fourier-tail consequence for triangular arrays of independent forest components. Every boundary-conditioned partition function and descendant subtree used below is obtained by deleting vertices from the original forest. The two results are the pointwise envelope (A.3) and the aggregate tail estimate (A.5).

The proof of the pointwise envelope has two principal stages. Lemma A.8 separates the contraction along a decorated path into a side-forest modulus–phase estimate, phase control from two consecutive rows, and a summation of relative decrements. Lemma A.9 then embeds that path estimate in a recursive maximal-modulus decomposition. Lemma A.10 records termination, disjointness, and the absence of repeated losses in the final strong induction.

A.1. Statement and notation

For a finite forest F , let

$$I_F(w) = \sum_{J \subseteq V(F)} w^{|J|} \mathbf{1}_{\{J \text{ is independent in } F\}}$$

be its independence polynomial. At activity $z > 0$, the hard-core law is

$$\Pr_{F,z}(J) = \frac{z^{|J|}}{I_F(z)} \mathbf{1}_{\{J \text{ is independent in } F\}}. \quad (\text{A.1})$$

Write $X_{F,z} = |J|$, $\mu_{F,z} = \mathbb{E} X_{F,z}$, and $V_{F,z} = \text{Var } X_{F,z}$. Its centered characteristic function, modulus, and logarithmic loss are

$$\phi_{F,z}(\theta) = \mathbb{E} e^{i\theta(X_{F,z} - \mu_{F,z})}, \quad M_{F,z}(\theta) = |\phi_{F,z}(\theta)|, \quad \ell_{F,z}(\theta) = -\log M_{F,z}(\theta), \quad (\text{A.2})$$

where $-\log 0 = +\infty$. Centering does not change the modulus, and hence

$$M_{F,z}(\theta) = \frac{|I_F(z e^{i\theta})|}{I_F(z)}.$$

When F is empty, $V_{F,z} = 0$, $M_{F,z} = 1$, and all assertions below hold trivially.

Theorem A.1 (full-domain Fourier envelope). For every $Z < \infty$ with $Z > 0$, there are constants $c_Z > 0$ and $\alpha_Z \in (0, 1)$ such that, for every finite forest F , every $0 < z \leq Z$, and every $|\theta| \leq \pi$,

$$\ell_{F,z}(\theta) \geq c_Z \min \{V_{F,z}\theta^2, (V_{F,z}\theta^2)^{\alpha_Z}\}. \quad (\text{A.3})$$

In particular, after changing c_Z ,

$$\ell_{F,z}(\theta) \geq c_Z \log(1 + V_{F,z}\theta^2). \quad (\text{A.4})$$

The following consequence is not needed in the main proof, but records the corresponding aggregate tail estimate.

Corollary A.2 (weighted aggregate Fourier tail). For each n , let $X_{n,1}, \dots, X_{n,m_n}$ be independent hard-core counts on finite forests $F_{n,1}, \dots, F_{n,m_n}$, all at a common activity $0 < z_n \leq Z$. Put

$$v_{n,j} = \text{Var } X_{n,j}, \quad W_n = \sum_j v_{n,j},$$

and let

$$\Phi_n(\theta) = \prod_j \phi_{F_{n,j}, z_n}(\theta)$$

be the centered characteristic function of their sum. If $W_n \rightarrow \infty$, then

$$\lim_{R \rightarrow \infty} \limsup_{n \rightarrow \infty} \int_{R \leq |u| \leq \pi \sqrt{W_n}} (1 + u^2) \left| \Phi_n \left(\frac{u}{\sqrt{W_n}} \right) \right| du = 0. \quad (\text{A.5})$$

The rest of the appendix proves the theorem and its corollary.

A.2. Rooted martingale increments and a uniform fourth moment

Root every tree component of F . For a vertex v , let T_v be the full descendant subtree rooted at v , and let $\text{ch}(v)$ be its children. Conditional on the parent of v being unoccupied, define

$$b_v = \Pr(v \text{ is occupied}), \quad q_v = 1 - b_v.$$

The root of a component is understood to have a virtual unoccupied parent. Deleting the root of T_v gives the exact recursion

$$\frac{b_v}{q_v} = z \prod_{u \in \text{ch}(v)} q_u. \quad (\text{A.7})$$

Consequently, with

$$\underline{q}_Z = (1 + Z)^{-1}, \quad \bar{p}_Z = \frac{Z}{1 + Z} = 1 - \underline{q}_Z,$$

one has, uniformly in F, z , and v ,

$$q_v \geq \underline{q}_Z, \quad b_v \leq \bar{p}_Z < 1. \quad (\text{A.8})$$

Let m_v^0 and m_v^1 be the mean occupation count in T_v , conditional respectively on v being absent and occupied, and put

$$d_v = m_v^1 - m_v^0.$$

Conditioning on v , and then on the child roots, gives

$$d_v = 1 - \sum_{u \in \text{ch}(v)} b_u d_u. \quad (\text{A.9})$$

The following estimate is the moment input for the central Fourier range.

Lemma A.3 (uniform fourth moment). For every finite Z there is $C_Z^{(4)} < \infty$ such that every finite forest and every $0 < z \leq Z$ satisfy

$$\mathbb{E}|X_{F,z} - \mu_{F,z}|^4 \leq C_Z^{(4)} V_{F,z} (1 + V_{F,z}). \quad (\text{A.10})$$

Proof. Introduce independent Bernoulli variables B_v , one at each vertex, with $\Pr(B_v = 1) = b_v$. Starting at the component roots, define the availability indicator R_v and the occupation indicator ξ_v by

$$R_v = 1 \quad \text{at a component root,} \quad R_v = 1 - \xi_{\text{par}(v)} \quad \text{otherwise,} \quad \xi_v = R_v B_v.$$

Induction from the leaves, using (A.7), shows that $(\xi_v)_{v \in F}$ has exactly the hard-core law (A.1). Order the vertices so that each parent precedes its children and reveal the B_v 's in that order. The corresponding Doob increment is

$$D_v = R_v d_v (B_v - b_v). \quad (\text{A.11})$$

Indeed, if $R_v = 0$, the Bernoulli variable B_v has no effect. If $R_v = 1$, changing B_v changes the conditional expected count in T_v from m_v^0 to m_v^1 , a difference of d_v . Thus

$$X_{F,z} - \mu_{F,z} = \sum_{v \in F} D_v. \quad (\text{A.12})$$

Set

$$e_v = b_v q_v d_v^2, \quad A = \sum_v e_v, \quad w_v = \mathbb{E} R_v.$$

Every component root has $w_v = 1$. At a nonroot vertex, its parent has marginal occupation probability at most $\bar{p}_Z$, so $w_v \geq \underline{q}_Z$. Orthogonality of the martingale increments therefore gives the exact variance identity

$$V_{F,z} = \sum_v w_v e_v, \quad \underline{q}_Z A \leq V_{F,z} \leq A. \quad (\text{A.13})$$

The predictable quadratic sum is

$$H = \sum_v \mathbb{E}(D_v^2 \mid \mathcal{F}_{v-}) = \sum_v R_v e_v.$$

Since $0 \leq H \leq A$ and $\mathbb{E}H = V_{F,z}$,

$$\mathbb{E}H^2 \leq A \mathbb{E}H \leq \underline{q}_Z^{-1} V_{F,z}^2. \quad (\text{A.14})$$

It remains to control the fourth moments of the individual increments. For each vertex set

$$s_v = \sum_{u \in \text{ch}(v)} b_u, \quad t_v = \sum_{u \in \text{ch}(v)} b_u d_u^2, \quad r_v = \sum_{u \in \text{ch}(v)} b_u d_u.$$

From (A.7), $q_u \leq e^{-b_u}$, and (A.8),

$$b_v \leq Z e^{-s_v}, \quad t_v \leq \underline{q}_Z^{-1} \sum_{u \in \text{ch}(v)} e_u, \quad r_v^2 \leq s_v t_v. \quad (\text{A.15})$$

If $|d_v| \leq 2$, then $b_v q_v d_v^4 \leq 4e_v$. If $|d_v| > 2$, (A.9) implies $|r_v| \geq |d_v|/2$, and hence

$$\begin{aligned} b_v q_v d_v^4 &\leq 16 b_v r_v^4 \leq 16 Z e^{-s_v} s_v^2 t_v^2 \\ &\leq 64 Z e^{-2} \underline{q}_Z^{-2} \left(\sum_{u \in \text{ch}(v)} e_u \right)^2. \end{aligned} \quad (\text{A.16})$$

Here $\sup_{s \geq 0} s^2 e^{-s} = 4e^{-2}$. Child sets belonging to distinct vertices are disjoint, so, with $C_A = 64 Z e^{-2} \underline{q}_Z^{-2}$,

$$\sum_v b_v q_v d_v^4 \leq 4A + C_A A^2. \quad (\text{A.17})$$

For completeness, we record the finite martingale estimate used to combine these bounds. If $D_1, \dots, D_N$ are real martingale differences,

$$h_k = \mathbb{E}(D_k^2 \mid \mathcal{F}_{k-1}), \quad H = \sum_k h_k,$$

then

$$\mathbb{E} \left| \sum_k D_k \right|^4 \leq \left(\frac{128}{9} \right)^2 \mathbb{E} H^2 + 6 \sum_k \mathbb{E} D_k^4. \quad (\text{A.18})$$

To prove (A.18), write $S_k = \sum_{j \leq k} D_j$ and $c_k = \mathbb{E}(D_k^3 \mid \mathcal{F}_{k-1})$. Conditional expansion and $|c_k| \leq (h_k \mathbb{E}(D_k^4 \mid \mathcal{F}_{k-1}))^{1/2}$ give

$$\mathbb{E}(S_k^4 - S_{k-1}^4 \mid \mathcal{F}_{k-1}) \leq 8 S_{k-1}^2 h_k + 3 \mathbb{E}(D_k^4 \mid \mathcal{F}_{k-1}).$$

After summation, if $S_* = \max_{k \leq N} |S_k|$ and $G = \sum_k \mathbb{E} D_k^4$, Cauchy–Schwarz gives

$$\mathbb{E} S_N^4 \leq 8 \mathbb{E}(S_*^2 H) + 3G \leq 8(\mathbb{E} S_*^4)^{1/2} (\mathbb{E} H^2)^{1/2} + 3G.$$

We use the finite Doob inequality

$$\mathbb{E} \max_{k \leq N} |S_k|^4 \leq \left(\frac{4}{3} \right)^4 \mathbb{E} |S_N|^4$$

whose proof is included next. Apply the first-crossing argument to the nonnegative submartingale $|S_k|$. For $\lambda > 0$, its first-crossing time gives

$$\lambda \Pr\left(\max_k |S_k| > \lambda\right) \leq \mathbb{E}\left[|S_N| \mathbf{1}_{\{\max_k |S_k| > \lambda\}}\right].$$

Integrating this inequality and using Hölder's inequality proves the displayed maximal estimate: multiplication by $4\lambda^2$, integration over $\lambda > 0$, and Fubini's theorem give

$$\mathbb{E} \max_{k \leq N} |S_k|^4 \leq \frac{4}{3} \mathbb{E}\left[|S_N| \max_{k \leq N} |S_k|^3\right].$$

Hölder's inequality then gives

$$\left(\mathbb{E} \max_{k \leq N} |S_k|^4\right)^{1/4} \leq \frac{4}{3} (\mathbb{E}|S_N|^4)^{1/4}.$$

Substitution in the bound for $\mathbb{E}S_N^4$ gives

$$\mathbb{E}S_N^4 \leq \frac{128}{9} (\mathbb{E}S_N^4)^{1/2} (\mathbb{E}H^2)^{1/2} + 3G.$$

Applying $ab \leq (a^2 + b^2)/2$ to the product on the right and moving $\frac{1}{2}\mathbb{E}S_N^4$ to the left proves (A.18).

For a Bernoulli variable of mean b ,

$$\mathbb{E}(B - b)^4 = b(1 - b)(b^3 + (1 - b)^3) \leq b(1 - b).$$

Thus (A.17) bounds $\sum_v \mathbb{E}D_v^4$. Combining (A.13), (A.14), (A.17), and (A.18) gives

$$\mathbb{E}|X_{F,z} - \mu_{F,z}|^4 \leq \frac{24}{q_Z} V_{F,z} + \left\{ \frac{1}{q_Z} \left(\frac{128}{9} \right)^2 + \frac{6C_A}{q_Z^2} \right\} V_{F,z}^2,$$

which is (A.10). Every constant depends only on Z . $\square$

A.3. Characteristic-function bounds

We next prove a full-frequency estimate which rules out exact or approximate lattice concentration at any fixed nonzero frequency.

Lemma A.4 (uniform characteristic-function gap). Let

$$\kappa_Z = \min \left\{ \frac{1 - e^{-7/16}}{8}, \frac{1}{8(1 + Z)} \right\}.$$

For every finite forest F , $0 < z \leq Z$, and $|\theta| \leq \pi$,

$$\ell_{F,z}(\theta) \geq \kappa_Z \min\{V_{F,z}, 1\} \sin^2(\theta/2). \quad (\text{A.19})$$

Proof. A forest is bipartite; fix a bipartition $V(F) = L \sqcup R$. Put

$$p = \frac{z}{1+z}, \quad q = p(1-p), \quad h = \sin^2(\theta/2), \quad a = 1-p+pe^{i\theta}, \quad r = |a| = \sqrt{1-4qh}.$$

The empty forest was disposed of in Section A.1, so assume F is nonempty.

For an independent set J , let

$$A_L(J) = |\{x \in L : N(x) \cap J = \emptyset\}|,$$

and define A_R analogously. Conditional on $J \cap R$, the available vertices in L are independent Bernoulli variables of mean p . Hence

$$|\phi_{F,z}(\theta)| \leq \mathbb{E}r^{A_L}, \quad |\phi_{F,z}(\theta)| \leq \mathbb{E}r^{A_R}. \quad (\text{A.20})$$

Suppose first that $p \geq 1/8$. Pointwise $A_L + A_R \geq 1$, and for nonnegative integers a_0, b_0 with $a_0 + b_0 \geq 1$, $r^{a_0} + r^{b_0} \leq 1 + r$. Averaging the two bounds in (A.20) gives

$$|\phi_{F,z}(\theta)| \leq \frac{1+r}{2} \leq 1 - p(1-p)h \leq \exp\left\{-\frac{h}{8(1+Z)}\right\}. \quad (\text{A.21})$$

This is stronger than (A.19) in this case.

Now suppose that $p \leq 1/8$, and put $n = |V(F)|$, $A = A_L + A_R$, and $U = n - A$. Every unavailable vertex has an occupied neighbor, so

$$U \leq \sum_{x \in J} \deg(x).$$

For every vertex x , deletion of x gives

$$\Pr(x \in J) = \frac{zI_{F-N[x]}(z)}{I_{F-x}(z) + zI_{F-N[x]}(z)} \leq p.$$

Since a forest has at most n edges, $\mathbb{E}U \leq 2pn$. Markov's inequality therefore gives $\Pr(A \geq n/2) \geq 1/2$. On this event at least one of A_L, A_R is at least $n/4$; consequently one fixed side $S \in \{L, R\}$ satisfies

$$\Pr(A_S \geq n/4) \geq \frac{1}{4}.$$

Using this event in (A.20), rather than applying Jensen's inequality, yields

$$|\phi_{F,z}(\theta)| \leq 1 - \frac{1}{4}(1 - r^{n/4}). \quad (\text{A.22})$$

Since $r \leq e^{-2p(1-p)h}$, if $t = np$, then

$$r^{n/4} \leq e^{-7th/16}.$$

Concavity of $1 - e^{-7x/16}$ on $[0, 1]$, followed by monotonicity when $t \geq 1$, gives

$$1 - e^{-7th/16} \geq (1 - e^{-7/16}) \min\{t, 1\}h. \quad (\text{A.23})$$

It remains to compare t with the variance. For distinct vertices x, y , either they are adjacent and cannot both be occupied, or conditioning on $x \in J$ leaves a hard-core law on $F - N[x]$, in which the marginal occupation probability of y is at most p . Thus

$$V_{F,z} \leq \mathbb{E}X_{F,z}^2 \leq np + n(n-1)p^2 \leq t + t^2.$$

It follows that

$$\min\{t, 1\} \geq \frac{1}{2} \min\{V_{F,z}, 1\}. \quad (\text{A.24})$$

Combining (A.22)–(A.24) with $1 - y \leq e^{-y}$ proves (A.19) in the low- p case. $\square$

The fourth-moment estimate and Lemma A.4 now give the small-frequency curvature needed later.

Lemma A.5 (small-frequency curvature). For every finite Z there are $x_Z^c > 0$ and $c_Z^c > 0$ such that

$$V_{F,z}\theta^2 \leq x_Z^c, \quad |\theta| \leq \pi \quad \implies \quad \ell_{F,z}(\theta) \geq c_Z^c V_{F,z}\theta^2 \quad (\text{A.25})$$

for every finite forest and every $0 < z \leq Z$.

Proof. Put $V = V_{F,z}$, $x = V\theta^2$, and let X' be an independent copy of $X = X_{F,z}$. If $V \leq 1$, Lemma A.4 and $\sin^2(\theta/2) \geq \theta^2/\pi^2$ give

$$\ell_{F,z}(\theta) \geq \frac{\kappa_Z}{\pi^2} x.$$

Suppose $V \geq 1$. With $Y = X - X'$, Lemma A.3 gives

$$\mathbb{E}Y^4 = 2\mathbb{E}|X - \mathbb{E}X|^4 + 6V^2 \leq (4C_Z^{(4)} + 6)V^2.$$

Because $1 - \cos t \geq t^2/2 - t^4/24$ for every real t ,

$$\begin{aligned} 1 - M_{F,z}(\theta)^2 &= \mathbb{E}[1 - \cos(\theta Y)] \\ &\geq x - \frac{4C_Z^{(4)} + 6}{24} x^2. \end{aligned} \quad (\text{A.26})$$

Choose

$$x_Z^c = \min \left\{ 1, \frac{12}{4C_Z^{(4)} + 6} \right\}.$$

Then the right side of (A.26) is at least $x/2$. Finally, $-\log r \geq (1 - r^2)/2$ for $0 < r \leq 1$, so $\ell_{F,z}(\theta) \geq x/4$. Taking

$$c_Z^c = \min \left\{ \frac{1}{4}, \frac{\kappa_Z}{\pi^2} \right\}$$

proves the assertion. $\square$

A.4. Root variance comparison and bounded-scale compactness

We first collect the rooted variance identities. Let T be a rooted tree, let r be its root, and set

$$Q = T - r, \quad R = T - N[r].$$

Thus Q is the forest of descendant subtrees rooted at the children of r , and R is the forest obtained by also deleting every child root. Write $b = b_r$, $q = 1 - b$, and let $d = d_r$. From this point through Section A.6, an omitted activity subscript always means the same fixed activity z ; for example, $V_T = V_{T,z}$, $M_T = M_{T,z}(\theta)$, and $\ell_T = \ell_{T,z}(\theta)$. Conditional variance at the root gives

$$V_T = qV_Q + bV_R + bq d^2. \quad (\text{A.27})$$

For the child roots u , put

$$E = \sum_u b_u q_u d_u^2, \quad R_0 = \sum_u \frac{b_u}{q_u}.$$

Weighted Cauchy–Schwarz and (A.9) give $(1 - d)^2 \leq R_0 E$. Moreover,

$$bR_0 \leq Z, \quad (\text{A.28})$$

because

$$\left(\prod_u q_u \right) \sum_u \frac{b_u}{q_u} = \sum_u b_u \prod_{w \neq u} q_w \leq 1$$

and $b/q = z \prod_u q_u$. Since $E \leq V_Q$, (A.27)–(A.28) imply, with

$$A_Z = 1 + 3Z, \quad B_Z = 1 + Z,$$

the depth-free root estimates

$$qV_Q \leq V_T \leq A_Z V_Q + 2b, \quad V_R \leq B_Z V_Q. \quad (\text{A.29})$$

Indeed, $bq d^2 \leq 2b + 2ZV_Q$, while $bV_R \leq ZV_Q$. Applying the first upper bound in (A.29) to every child subtree and summing also gives

$$V_Q \leq A_Z V_R + 2L, \quad L = \log \frac{I_Q(z)}{I_R(z)} = \sum_u -\log q_u. \quad (\text{A.30})$$

The next lemma is the bounded-scale compactness statement required in the global induction. Its proof is included because ordinary compactness of a fixed family of polynomials would not be uniform in forest size or depth.

Lemma A.6 (bounded-scale compactness). For every $Z < \infty$ and every $0 < a \leq b < \infty$, there is $\varepsilon = \varepsilon(Z, a, b) > 0$ such that

$$a \leq V_{F,z} \theta^2 \leq b, \quad |\theta| \leq \pi \quad \implies \quad \ell_{F,z}(\theta) \geq \varepsilon \quad (\text{A.31})$$

for every finite forest and every $0 < z \leq Z$.

Proof. We first prove a one-edge modulus transfer. Root a tree T , and let $U = T_v$ be any descendant subtree. If $U \neq T$, let s be the parent of its root. Under the hard-core law on T ,

$$w := \Pr(s \text{ is unoccupied}) = \frac{I_{T-s}(z)}{I_T(z)} \geq \frac{1}{1+z} \geq \underline{q}_Z, \quad (\text{A.32})$$

because $T - N[s]$ is an induced subgraph of $T - s$. Conditional on this event, the law on U is its free hard-core law and is independent of the outside configuration. Splitting the characteristic function according to the event and using the triangle inequality yields

$$1 - M_{U,z}(\theta) \leq \underline{q}_Z^{-1}(1 - M_{T,z}(\theta)). \quad (\text{A.33})$$

Only the immediate outside endpoint is conditioned; no factor depending on the depth of U appears.

Suppose that (A.31) fails. There are F_j, z_j, θ_j in the stated range such that

$$a \leq \theta_j^2 V_{F_j, z_j} \leq b, \quad M_{F_j, z_j}(\theta_j) \longrightarrow 1. \quad (\text{A.34})$$

After passing to a subsequence, either $|\theta_j| \geq \delta > 0$, in which case Lemma A.4 gives a fixed positive loss, or $\theta_j \rightarrow 0$. We treat the latter case.

Write F_j as the disjoint union of its tree components and put $x_{j,i} = \theta_j^2 V_{j,i}$. By Lemma A.5 and additivity of logarithmic loss,

$$\sum_{i: x_{j,i} \leq x_Z^c} x_{j,i} \longrightarrow 0.$$

There are at most b/x_Z^c remaining components. Hence one can select a tree component T_j and a constant $a_1 > 0$, depending only on Z, a, b , so that

$$a_1 \leq \theta_j^2 V_{T_j, z_j} \leq b, \quad M_{T_j, z_j}(\theta_j) \longrightarrow 1. \quad (\text{A.35})$$

Set $r_0 = \min\{x_Z^c, a_1/2\}$. We claim that, for all large j , every descendant subtree $U \subseteq T_j$ with

$$\theta_j^2 V_{U, z_j} > r_0 \quad (\text{A.36})$$

has a descendant subtree rooted at one of its children that satisfies the same inequality. Let $Q_U = U - \text{root}(U)$. By (A.29), uniformly in U ,

$$\theta_j^2 V_{Q_U, z_j} \geq \frac{r_0}{2A_Z} \quad (\text{A.37})$$

once $\theta_j^2 \leq r_0/4$. Conditioning the root of U to be absent, and then applying (A.33) directly from T_j to U , gives

$$1 - M_{Q_U, z_j}(\theta_j) \leq \underline{q}_Z^{-2}(1 - M_{T_j, z_j}(\theta_j)) \rightarrow 0, \quad (\text{A.38})$$

uniformly over all such U . If every child component W of Q_U had $\theta_j^2 V_W \leq r_0$, Lemma A.5 and independence would give

$$\ell_{Q_U, z_j}(\theta_j) \geq c_Z^c \theta_j^2 V_{Q_U, z_j} \geq \frac{c_Z^c r_0}{2A_Z},$$

contradicting (A.38). The claim follows.

Starting from (A.35), the claim produces a strict descending chain of full descendant subtrees inside the same finite tree T_j , each satisfying (A.36). The chain must end at a leaf. A one-vertex tree has variance at most $1/4$, so its scaled variance is less than r_0 for all large j , a contradiction. Thus (A.31) holds. $\square$

Combining Lemmas A.5 and A.6, for every fixed $X_0 < \infty$ and every $\alpha \in (0, 1)$, there is $c = c(Z, X_0, \alpha) > 0$ such that

$$\ell_{F, z}(\theta) \geq c H_\alpha(V_{F, z} \theta^2), \quad V_{F, z} \theta^2 \leq X_0, \quad H_\alpha(x) = \min\{x, x^\alpha\}. \quad (\text{A.39})$$

A.5. Variance and Fourier contraction along a path

We now establish the depth-free estimates used when the scaled variance is unbounded.

Lemma A.7 (variance bound along a path). Let $v_0, v_1, \dots, v_L$, $L \geq 1$, be a downward path in a rooted tree, and let S_k be the set of children of v_k other than v_{k+1} . Let $\mathcal{B}_k = \bigsqcup_{u \in S_k} T_u$ be their side forest. Then

$$V_{T_{v_0}, z} \leq C_Z \left\{ \sum_{k=0}^{L-1} b_{v_k} + V_{T_{v_L}, z} + \sum_{k=0}^{L-1} V_{\mathcal{B}_k, z} \right\}, \quad (\text{A.40})$$

where C_Z is independent of L .

Proof. Write $b_k = b_{v_k}$, $d_k = d_{v_k}$, $e_k = b_k q_k d_k^2$, and

$$c_k = \sum_{u \in S_k} b_u d_u, \quad E_k = \sum_{u \in S_k} b_u q_u d_u^2, \quad P_0 = 1, \quad P_k = \prod_{r=1}^k b_r.$$

Iteration of $d_k = 1 - b_{k+1} d_{k+1} - c_k$ gives

$$|d_0| \leq \sum_{k < L} P_k + \sum_{k < L} P_k |c_k| + P_L |d_L|.$$

The first sum is at most $\underline{q}_Z^{-1}$. If $R_k = \sum_{u \in S_k} b_u / q_u$, then $c_k^2 \leq R_k E_k$, while the occupation-probability identity gives $b_k R_k \leq Z$, exactly as in (A.28). A second weighted Cauchy-Schwarz inequality gives

$$\left(\sum_{k < L} P_k |c_k| \right)^2 \leq \left(\sum_{k < L} P_k \right) \left(\sum_{k < L} P_k R_k E_k \right) \leq \underline{q}_Z^{-1} \sum_{k < L} P_k R_k E_k.$$

For $k = 0$, $b_0 P_0 R_0 \leq Z$. For $k \geq 1$,

$$b_0 P_k R_k = b_0 P_{k-1} b_k R_k \leq Z \bar{p}_Z^{k-1}.$$

The bottom term satisfies

$$b_0 P_L^2 d_L^2 = b_0 P_{L-1}^2 \frac{b_L}{q_L} e_L \leq Z \bar{p}_Z^{2L-1} e_L.$$

Using $e_0 \leq b_0 d_0^2$ and $(r + s + t)^2 \leq 3(r^2 + s^2 + t^2)$, we obtain

$$e_0 \leq C_Z \left\{ b_0 + \sum_{k < L} \bar{p}_Z^{\max\{k-1, 0\}} E_k + \bar{p}_Z^{2L-1} e_L \right\}. \quad (\text{A.41})$$

Apply (A.41) to every suffix $v_r, \dots, v_L$. Each E_k receives total weight at most

$$2 + \sum_{j \geq 1} \bar{p}_Z^j < \infty,$$

and the bottom term receives another bounded geometric weight. Finally, the variance identity (A.13), applied inside each disjoint side or bottom descendant subtree, bounds the sum of its e_v 's by $\bar{p}_Z^{-1}$ times its variance. Adding the path, side, and bottom contributions proves (A.40). $\square$

We next prove the Fourier analogue of (A.40). Let $v_0, \dots, v_L$ be a downward path. At v_k , let $\mathcal{S}_k$ be the side forest, and write $\text{Comp}(\mathcal{S}_k)$ for its set of tree components. Define

$$\begin{aligned} A_k(w) &= I_{\mathcal{S}_k}(w), & C_k(w) &= \prod_{U \in \text{Comp}(\mathcal{S}_k)} I_{U-\text{root}(U)}(w), \\ a_k &= \frac{A_k(ze^{i\theta})}{A_k(z)}, & c_k &= \frac{C_k(ze^{i\theta})}{C_k(z)}, & \Lambda_k &= -\log \prod_{U \in \text{Comp}(\mathcal{S}_k)} (1 - b_U). \end{aligned} \quad (\text{A.42})$$

Here b_U is the occupation probability of the root of U in its free law. Empty products equal one. We also write $b_k = b_{v_k}$, $q_k = 1 - b_k$, and $e_1 = (1, 0)^\top$. When $a_k c_k \neq 0$, define

$$\gamma_k = \arg(e^{i\theta} c_k / a_k) \quad \text{in } \mathbb{R}/(2\pi\mathbb{Z}).$$

If $a_k c_k = 0$, set $\gamma_k = 0$; in that case every later occurrence of its phase term is bounded by the corresponding radial defect and the chosen value is immaterial.

Lemma A.8 (transfer-matrix contraction along a decorated path). There are constants $c_Z, C_Z > 0$ such that, for every path as above and every complex bottom vector $f_L \in \mathbb{C}^2$ whose coordinate moduli are at most one,

$$|e_1^\top K_0 K_1 \cdots K_{L-1} f_L| \leq C_Z \exp \left\{ -c_Z \sin^2(\theta/2) \sum_{k < L} (b_k + \Lambda_k) \right\}, \quad (\text{A.43})$$

where

$$K_k = \begin{pmatrix} q_k a_k & b_k e^{i\theta} c_k \\ a_k & 0 \end{pmatrix}. \quad (\text{A.44})$$

The constants are independent of the path length, the side forests, and the bottom subtree.

Proof. We first prove the phase restriction imposed by a side forest. Attach a new root to the roots of the components of an arbitrary side forest S , and let $\bar{b}$ and $\bar{q} = 1 - \bar{b}$ be the new root's occupation and vacancy probabilities. With

$$a = \frac{I_S(z e^{i\theta})}{I_S(z)}, \quad c = \frac{\prod_U I_{U-\text{root}(U)}(z e^{i\theta})}{\prod_U I_{U-\text{root}(U)}(z)},$$

the normalized uncentered characteristic function of the resulting tree is

$$\bar{q}a + \bar{b}e^{i\theta}c. \quad (\text{A.45})$$

Put $h = \sin^2(\theta/2)$, $r = |a|$, and $s = |c|$. When $a, c \neq 0$, let

$$\gamma = \arg(e^{i\theta}c/a) \quad \text{in } \mathbb{R}/(2\pi\mathbb{Z}),$$

and set $\gamma = 0$ otherwise.

Sublemma A.8.1 (side-forest radial and phase coercivity). With the quantities above, the radial defects and the relative phase satisfy (A.48). When $a = 0$ or $c = 0$, the phase term is omitted and the corresponding radial defect controls the estimate.

Proof.

Its variance is bounded below by $\underline{q}_Z^2 \bar{b}$. To see this, use the notation d, E, R_0 preceding (A.28). The root and child-root martingale increments in (A.13) give

$$V \geq \bar{b}\bar{q}d^2 + \underline{q}_Z E \geq \underline{q}_Z \bar{b} \left(d^2 + \frac{1}{Z}(1-d)^2 \right) \geq \underline{q}_Z^2 \bar{b}. \quad (\text{A.46})$$

The childless case follows directly from $V = \bar{b}\bar{q} \geq \underline{q}_Z \bar{b}$. Lemma A.4 and (A.46) imply

$$1 - |\bar{q}a + \bar{b}e^{i\theta}c| \geq c_Z \bar{b}h. \quad (\text{A.47})$$

If $\text{dist}(\gamma, 2\pi\mathbb{Z})$ denotes its representative in $[0, \pi]$, rationalizing the triangle deficit gives

$$\begin{aligned} 1 - |\bar{q}a + \bar{b}e^{i\theta}c| &\leq \bar{q}(1-r) + \bar{b}(1-s) \\ &\quad + \frac{\bar{b}}{2} \text{dist}(\gamma, 2\pi\mathbb{Z})^2. \end{aligned}$$

When $a = 0$ or $c = 0$, the corresponding radial defect controls the whole estimate. Combining this bound with (A.47) gives the side constraint

$$\bar{b}h \leq C_Z \{ (1-r) + \bar{b}(1-s) + \bar{b} \text{dist}(\gamma, 2\pi\mathbb{Z})^2 \}. \quad (\text{A.48})$$

□

We now turn to the matrix product. The polynomial recursions along the path are

$$\begin{aligned} I_{T_{v_k}}(w) &= A_k(w)I_{T_{v_{k+1}}}(w) + wC_k(w)I_{T_{v_{k+1}-v_k}}(w), \\ I_{T_{v_k-v_k}}(w) &= A_k(w)I_{T_{v_{k+1}}}(w). \end{aligned}$$

After normalization at $w = z$, these recursions are exactly (A.44). Coefficient positivity gives the coordinatewise bound on every bottom vector.

Starting with $(x_0, y_0) = (1, 0)$, set

$$(x_{k+1}, y_{k+1}) = (x_k, y_k)K_k, \quad N_k = |x_k| + |y_k|.$$

Then N_k is nonincreasing. More precisely, if

$$\Delta_k = N_k - N_{k+1}, \quad D_k = q_k|x_k| + |y_k| - |q_kx_k + y_k|,$$

then

$$\Delta_k = D_k + (1 - |a_k|)|q_kx_k + y_k| + b_k(1 - |c_k|)|x_k|. \quad (\text{A.49})$$

Sublemma A.8.2 (phase control from two consecutive rows). Whenever $k+1 < L$, the decrements in two consecutive rows satisfy

$$\begin{aligned} |x_k|\{(1 - |a_k|) + b_k(1 - |c_k|) + b_k \operatorname{dist}(\gamma_k, 2\pi\mathbb{Z})^2\} \\ \leq C_Z(\Delta_k + \Delta_{k+1}), \end{aligned} \quad (\text{A.50})$$

Proof. For $A, B \geq 0$,

$$A + B - |Ae^{iu} + Be^{iv}| \geq \frac{2AB}{\pi^2(A+B)} \operatorname{dist}(u - v, 2\pi\mathbb{Z})^2. \quad (\text{A.51})$$

Put $S = q_kx_k + y_k$ and $X = |x_k|$. If $|S| < q_kX/2$, then $D_k \geq q_kX/2 \geq \underline{q}_Z X/2$, which controls every term on the left of (A.50). Otherwise the two radial terms in (A.49) control the first two terms of (A.50). If either $|a_k| < 1/2$ or $|c_k| < 1/2$, a radial term also controls the phase term. We may therefore assume $|a_k|, |c_k| \geq 1/2$. Then

$$|x_{k+1}| \geq \underline{q}_Z X/4, \quad |y_{k+1}| \geq b_k X/2.$$

Let α be the circular angle from x_k to S . A direct rationalization gives the required cost as follows. Put $A = q_kX$, $R = |S|$, and $Y = |y_k|$. If $Y = 0$ and $R = A$, then $S = q_kx_k$, so $\alpha = 0$, and the desired phase term vanishes. In every other case with $R \geq A$, the denominator below is positive and

$$D_k = A + Y - R = \frac{2AR(1 - \cos \alpha)}{Y + R - A} \geq A(1 - \cos \alpha),$$

because $Y + R - A \leq 2R$. If $A/2 < R < A$, then

$$D_k \geq Y - (A - R) = \frac{2AR(1 - \cos \alpha)}{Y + A - R} \geq R(1 - \cos \alpha).$$

If $R \leq A/2$, then $D_k \geq A - R \geq A/2$. Since $1 - \cos t \geq 2 \operatorname{dist}(t, 2\pi\mathbb{Z})^2/\pi^2$, these three cases give

$$D_k \geq c_Z X \operatorname{dist}(\alpha, 2\pi\mathbb{Z})^2.$$

Applying (A.51) at the next row gives

$$D_{k+1} \geq c_Z b_k X \operatorname{dist}(\beta, 2\pi\mathbb{Z})^2,$$

where β is the circular angle from x_{k+1} to y_{k+1} . Indeed, $q_{k+1}|x_{k+1}| \geq \underline{q}_Z^2 X/4$ and $b_k X/2 \leq |y_{k+1}| \leq b_k X$, so the harmonic coefficient in (A.51) is at least a Z -dependent multiple of $b_k X$. The row identities

$$x_{k+1} = a_k S, \quad y_{k+1} = b_k e^{i\theta} c_k x_k$$

show explicitly that

$$\begin{aligned} \beta &= \arg y_{k+1} - \arg x_{k+1} \\ &= (\theta + \arg c_k - \arg a_k) - (\arg S - \arg x_k) = \gamma_k - \alpha \pmod{2\pi}. \end{aligned}$$

The circular triangle inequality, followed by $(a+b)^2 \leq 2a^2 + 2b^2$, proves (A.50). All zero-vector cases were already covered by the radial terms. $\square$

Sublemma A.8.3 (summation of relative decrements). If $N_k > 0$ for every $k \leq L$, then the relative decrements control the total path occupation weight through (A.55).

Proof. The root probability $\bar{b}$ in (A.48) is formed without the distinguished child, whereas b_k includes it. Their odds differ by the distinguished child vacancy $q_{k+1} \in [\underline{q}_Z, 1]$, and therefore

$$\underline{q}_Z \bar{b} \leq b_k \leq \bar{b}.$$

Thus (A.48) and (A.50) imply

$$b_k h |x_k| \leq C_Z (\Delta_k + \Delta_{k+1}), \quad k < L-1. \quad (\text{A.52})$$

To remove the factor $|x_k|$, put

$$Q_k = \prod_{U \in \operatorname{Comp}(\mathcal{S}_k)} (1 - b_U).$$

The recursion for the positive occupation probabilities gives

$$b_{k+1} \leq z, \quad b_k \geq \underline{q}_Z^2 z Q_k. \quad (\text{A.53})$$

If $Q_k \geq 1/2$, then $b_{k+1} \leq C_Z b_k$. If $Q_k < 1/2$, then $\sum_{U \in \operatorname{Comp}(\mathcal{S}_k)} b_U > 1/2$, and (A.46) together with Lemma A.4, applied componentwise, gives

$$1 - |a_k| \geq c_Z h. \quad (\text{A.54})$$

Assume first that every $N_k > 0$, and set

$$\delta_k = \frac{N_k - N_{k+1}}{N_k}.$$

If $|x_k| \geq N_k/2$, (A.52) controls $b_k h$ through $C_Z(\delta_k + \delta_{k+1})$. If $|x_k| < N_k/2$, then $k \geq 1$ and

$$|y_k| = b_{k-1} |c_{k-1}| |x_{k-1}| > N_k/2.$$

If $N_{k-1} > 2N_k$, the single decrement $\delta_{k-1} > 1/2$ gives the bounded quantity $b_k h$. Otherwise $|x_{k-1}| \geq N_{k-1}/4$. Apply (A.53)–(A.54) at $k-1$: if $Q_{k-1} \geq 1/2$, (A.53) gives $b_k \leq 2\underline{q}_Z^{-2} b_{k-1}$, and (A.52) at $k-1$ gives

$$b_k h \leq C_Z(\delta_{k-1} + \delta_k).$$

Suppose instead that $Q_{k-1} < 1/2$, and put $S_{k-1} = q_{k-1} x_{k-1} + y_{k-1}$. By (A.54), $1 - |a_{k-1}| \geq c_Z h$. If $|S_{k-1}| \geq \underline{q}_Z N_{k-1}/8$, the radial decrement in (A.49) gives

$$\Delta_{k-1} \geq (1 - |a_{k-1}|) |S_{k-1}| \geq c_Z h N_{k-1}.$$

If $|S_{k-1}| < \underline{q}_Z N_{k-1}/8$, then

$$D_{k-1} \geq q_{k-1} |x_{k-1}| - |S_{k-1}| \geq \underline{q}_Z N_{k-1}/8.$$

Thus $\delta_{k-1} \geq c_Z h$ in either subcase, and this controls $b_k h$ because $b_k \leq 1$. The four cases above use the following relative decrements:

case at row k	controlling relative decrements
$ x_k \geq N_k/2$	$\delta_k + \delta_{k+1}$
$ x_k < N_k/2, N_{k-1} > 2N_k$	δ_{k-1}
$Q_{k-1} \geq 1/2$	$\delta_{k-1} + \delta_k$
$Q_{k-1} < 1/2$	δ_{k-1}

A fixed δ_j can therefore occur only in the estimates for rows $j-1, j, j+1$, so it is counted at most three times. The final unpaired row contributes at most one. Finally,

$$\sum_{k < L} \delta_k \leq \sum_{k < L} -\log \frac{N_{k+1}}{N_k} = \log \frac{1}{N_L},$$

because $1 - t \leq -\log t$. Summation gives

$$h \sum_{k < L} b_k \leq C_Z \left(1 + \log \frac{1}{N_L} \right). \quad (\text{A.55})$$

□

If some $N_k = 0$, every later norm vanishes and the desired estimate is immediate. Otherwise Sublemma A.8.3 yields

$$N_L \leq C_Z \exp \left\{ -c_Z h \sum_{k < L} b_k \right\}. \quad (\text{A.56})$$

It remains to include the side barriers Λ_k . For every component U of $\mathcal{S}_k$, the root calculation in (A.46), applied to U at its own root, gives

$$V_{U,z} \geq \underline{q}_Z^2 b_U, \quad \min\{V_{U,z}, 1\} \geq \underline{q}_Z^2 b_U.$$

Lemma A.4 and additivity of logarithmic loss over the independent components therefore give

$$-\log |a_k| \geq c_Z h \sum_{U \in \text{Comp}(\mathcal{S}_k)} b_U \geq c_Z h \Lambda_k, \quad (\text{A.57})$$

where the last inequality uses $-\log(1 - b_U) \leq b_U / \underline{q}_Z$. Put $r_k = |a_k|$. The full root odds are

$$\frac{b_k}{q_k} = z q_{k+1} e^{-\Lambda_k}.$$

There is $c_Z > 0$ such that, uniformly for $0 \leq h \leq 1$ and $\Lambda \geq 0$,

$$r_k q_k + b_k \leq e^{-c_Z h \Lambda_k}. \quad (\text{A.58})$$

For clarity, this scalar estimate requires no compactness theorem. Let $a_Z > 0$ be the constant in (A.57), and put $t = h\Lambda$. If $t \leq 1$, then (A.57) and $q_k \geq \underline{q}_Z$ give

$$1 - (r_k q_k + b_k) = q_k(1 - r_k) \geq \underline{q}_Z(1 - e^{-a_Z t}) \geq \underline{q}_Z(1 - e^{-a_Z})t.$$

If $t \geq 1$, the same bound gives $r_k q_k + b_k \leq 1 - \delta_Z$, where $\delta_Z = \underline{q}_Z(1 - e^{-a_Z}) > 0$. On the other hand,

$$r_k q_k + b_k \leq e^{-a_Z t} + Z e^{-\Lambda} \leq e^{-a_Z t} + Z e^{-t}.$$

Let $m_Z = \min\{a_Z, 1\}$, and choose $T_Z < \infty$ so that $(1 + Z)e^{-m_Z t} \leq e^{-m_Z t/2}$ for $t \geq T_Z$. On $[1, T_Z]$, $1 - \delta_Z \leq e^{-c_Z t}$ after choosing $c_Z \leq -T_Z^{-1} \log(1 - \delta_Z)$; for $t \geq T_Z$, use the last display and choose $c_Z \leq m_Z/2$. Together with the $t \leq 1$ estimate, this proves (A.58) with uniform constants.

Finally, (A.49) also gives

$$N_{k+1} \leq (r_k q_k + b_k) N_k.$$

Multiplying and using (A.58) yields exponential decay in $h \sum_k \Lambda_k$. Taking the geometric mean with (A.56) gives

$$N_L \leq C_Z \exp \left\{ -c_Z h \sum_{k < L} (b_k + \Lambda_k) \right\}.$$

The pairing with f_L is at most N_L , which proves (A.43). $\square$

A.6. Recursive decomposition along a maximal-modulus branch

Fix $z \in (0, Z]$, $|\theta| \leq \pi$, and an ambient finite rooted tree. For a nonempty descendant subtree $U = T_v$, define its scale and logarithmic Fourier loss by

$$x_U = \theta^2 V_{U,z}, \quad \ell_U = \ell_{U,z}(\theta).$$

The two deletion forests associated with U are $Q_U = U - v$ and $R_U = U - N_U[v]$, at the same z and θ . Every nonempty component of either forest is a strict descendant subtree. When the current subtree is denoted by T and its root by r , write $Q = T - r$ and $R = T - N[r]$. Choose $H \in \{Q, R\}$ so that

$$M_H(\theta) = \max\{M_Q(\theta), M_R(\theta)\}.$$

Ties may be resolved arbitrarily.

The normalized root decomposition and the triangle inequality give

$$M_T \leq qM_Q + bM_R \leq M_H, \quad \ell_T \geq \ell_H = \sum_{U \text{ component of } H} \ell_U. \quad (\text{A.59})$$

Fix a threshold

$$X_* > 4A_Z\pi^2.$$

For a subtree T with $x_T \geq X_*$, call its root **narrow** if $H = R$ and

$$x_R < \frac{x_Q}{2A_Z}. \quad (\text{A.60})$$

If the root is not narrow, (A.29) implies

$$x_H \geq \delta_Z x_T, \quad \delta_Z = (4A_Z^2)^{-1}. \quad (\text{A.61})$$

Indeed, $x_T \leq A_Z x_Q + 2\pi^2$ gives $x_Q \geq x_T/(2A_Z)$; this handles $H = Q$, and (A.60) handles the nonnarrow $H = R$ case.

Starting at a nonnarrow subtree, choose one component of H as the next subtree and place all other components in a discarded family $\mathcal{D}$. A Q -choice moves one edge down the rooted tree. An R -choice moves two edges through the child containing the selected grandchild component. Include that intermediate child in the path. After any finite number of choices, the resulting decomposition consists of the nested selected subtrees, the discarded components, and this vertex path. Let B be the final subtree, and define

$$\mathcal{K} = \theta^2 \left\{ \sum_{v \text{ on the path above } B} b_v + \sum_{R\text{-choices at } v} \log \frac{I_{Q_v}(z)}{I_{R_v}(z)} \right\}. \quad (\text{A.62})$$

Here $Q_v = T_v - v$ and $R_v = T_v - N[v]$ are the two deletion forests of the current descendant subtree T_v .

Lemma A.9 (maximal-modulus path decomposition). The path, final subtree B , and discarded family $\mathcal{D}$ above satisfy, simultaneously,

$$x_T \leq C_Z \left(\mathcal{K} + x_B + \sum_{U \in \mathcal{D}} x_U \right), \quad (\text{A.63})$$

$$\ell_T \geq \ell_B + \sum_{U \in \mathcal{D}} \ell_U, \quad (\text{A.64})$$

and

$$\ell_T \geq c_Z \mathcal{K} - C_Z. \quad (\text{A.65})$$

All members of $\mathcal{D}$, together with B , are pairwise disjoint descendant subtrees.

Proof. Apply Lemma A.7 to the filled path. At a Q -choice, every side subtree is a discarded component. At an R -choice through $v - u - w$, each side child subtree T_s , $s \neq u$, obeys (A.29):

$$V_{T_s} \leq A_Z \sum_{t \in \text{ch}(s)} V_{T_t} + 2b_s.$$

The T_t 's on the right, and the side children of u other than w , are discarded R -components. Also

$$\sum_{s \in \text{ch}(v)} b_s \leq \sum_s -\log(1 - b_s) = \log \frac{I_{Q_v}(z)}{I_{R_v}(z)}.$$

No descendant subtree is used at two different choices. Multiplying (A.40) by θ^2 proves (A.63).

At every choice, (A.59) gives the loss of the distinguished component plus all newly discarded components. Repeated substitution, rather than addition of inequalities containing repeated intermediate losses, proves (A.64).

For (A.65), apply Lemma A.8 to the filled path. At an R -choice through $v - u - w$,

$$\log \frac{I_{Q_v}(z)}{I_{R_v}(z)} = \sum_{s \neq u} -\log(1 - b_s) - \log(1 - b_u).$$

The first sum is the side barrier Λ_v , while $-\log(1 - b_u) \leq b_u/q_Z$ is bounded by the contribution of the next path vertex u , which lies above B . Finally, $\sin^2(\theta/2) \geq \theta^2/\pi^2$. Taking logarithms in (A.43) proves (A.65). Disjointness follows because all later discarded subtrees lie inside the preceding distinguished subtree, whereas earlier discarded subtrees lie outside it. $\square$

A high-scale narrow subtree already has a direct loss estimate. Indeed, summing (A.29) over its child subtrees and using (A.60) gives

$$\log \frac{I_Q(z)}{I_R(z)} \geq \sum_s b_s > \frac{V_Q}{4}.$$

We record directly the scalar one-row estimate needed here. Write

$$\Lambda = \log \frac{I_Q(z)}{I_R(z)}, \quad a = \frac{I_Q(ze^{i\theta})}{I_Q(z)}, \quad c = \frac{I_R(ze^{i\theta})}{I_R(z)}.$$

The root decomposition and the triangle inequality give

$$M_T(\theta) = |qa + be^{i\theta}c| \leq q|a| + b.$$

Moreover $b/q = ze^{-\Lambda}$, while (A.57) gives $|a| \leq \exp(-a_Z h \Lambda)$. The scalar argument used in the proof of (A.58), with no distinguished-child factor, therefore yields

$$M_T(\theta) \leq \exp(-c_Z h \Lambda).$$

For a narrow subtree, (A.30) and (A.60) imply

$$V_Q \leq A_Z V_R + 2\Lambda < \frac{V_Q}{2} + 2\Lambda, \quad \Lambda > \frac{V_Q}{4}.$$

The high-scale estimate immediately preceding (A.61) also gives $x_Q \geq x_T/(2A_Z)$. Since $h \geq \theta^2/\pi^2$, we conclude that

$$\ell_T \geq c_Z h \Lambda \geq c_Z \theta^2 V_Q \geq c_Z x_T.$$

After decreasing c_Z , this proves the stated, slightly weaker form

$$\ell_T \geq c_Z x_T - C_Z \tag{A.66}$$

for every high-scale narrow subtree.

We now complete the proof of Theorem A.1. Let

$$H_\alpha(x) = \min\{x, x^\alpha\}, \quad 0 < \alpha < 1.$$

This function is increasing and concave on $[0, \infty)$, with $H_\alpha(0) = 0$.

Let $C_1 \geq 1$ dominate both the constant in (A.63) and that constant multiplied by δ_Z^{-1} . Choose

$$0 < \sigma < \frac{1}{16C_1}, \quad a_0 = \frac{1}{2C_1}.$$

Choose a bounded-scale threshold $X_0 \geq X_*$, to be enlarged below, and fix a descendant subtree T of scale $X = x_T \geq X_0/\sigma$. We define one finite path decomposition and its recursive family only after removing the direct narrow case. If T is narrow, (A.66) applies and no recursive family is formed. Suppose henceforth that T is not narrow. At the current subtree U :

1. stop if $x_U \leq X_0$ or if U is narrow;
2. otherwise choose its deletion forest H_U by the preceding maximal-modulus rule; if exactly one component W of H_U satisfies $x_W > \sigma X$, put every other component of H_U into the discarded family and continue with W ;

3. stop if no component or at least two components of H_U have scale greater than σX .

Continue until the rule stops, let B be the final subtree, and let $\mathcal{D}$ be the union of all discarded component families. Define the recursive family $\mathcal{C}(T)$ by

$$\mathcal{C}(T) = \begin{cases} \mathcal{D} \cup \{B\}, & x_B \leq X_0 \text{ or } B \text{ is narrow,} \\ \mathcal{D} \cup \text{Comp}(H_B), & \text{otherwise.} \end{cases}$$

Lemma A.10 (recursive-decomposition bookkeeping). For every nonnarrow descendant subtree T to which the preceding construction is applied, the continuation terminates. The family $\mathcal{C}(T)$ consists of pairwise disjoint strict descendant subtrees, no intermediate logarithmic loss is counted more than once, and

$$\ell_T \geq \sum_{U \in \mathcal{C}(T)} \ell_U.$$

Moreover, the constant $C_1 = C_1(Z)$ chosen above satisfies (A.67).

Proof. Every continuation replaces the current subtree by a strict descendant subtree, so finiteness of T proves termination.

In the second case the stopping rule says that the final deletion forest has either no component above σX or at least two such components. Every member of $\mathcal{D}$ was discarded at a step having a unique component above σX , and hence has scale at most σX . The members of $\mathcal{C}(T)$ are pairwise disjoint strict descendant subtrees: discarded subtrees at an earlier step lie outside every later selected subtree, and the components of the final deletion forest are disjoint inside B . If B itself is included, then the selected path has positive length: the initial subtree was neither bounded-scale nor narrow, whereas retaining B means that a later subtree is bounded-scale or narrow.

After t continuation steps, let B_t be the distinguished component still being followed and let $\mathcal{D}_{\leq t}$ be all components discarded so far. Induction in (A.64) gives the invariant

$$\ell_T \geq \ell_{B_t} + \sum_{U \in \mathcal{D}_{\leq t}} \ell_U,$$

and the members of $\mathcal{D}_{\leq t}$, together with B_t , are pairwise disjoint. At the next step, (A.64) replaces the single term ℓ_{B_t} by the loss of the new distinguished component plus the newly discarded components. Hence no earlier loss remains to be counted again. At termination, (A.59) is used only if the final distinguished component is replaced by its deletion-forest components. This proves the exact loss comparison

$$\ell_T \geq \sum_{U \in \mathcal{C}(T)} \ell_U.$$

Likewise, (A.63) gives the required scale comparison when B is retained. When B is replaced by the components of its deletion forest, (A.61) gives $x_B \leq \delta_Z^{-1} x_{H_B}$, and forest variance additivity gives $x_{H_B} = \sum_{U \in \text{Comp}(H_B)} x_U$. Thus the chosen $C_1 = C_1(Z)$ satisfies in both cases

$$X \leq C_1 \left(\mathcal{K} + \sum_{U \in \mathcal{C}(T)} x_U \right). \quad (\text{A.67})$$

□

Except for a narrow endpoint with $x_B \geq \sigma X$, which is treated directly below, the stopping rule gives the following dichotomy for the child scales:

$$\begin{aligned} & x_i \leq \sigma X \quad \text{for every } i, \quad \text{or} \\ & \text{at least two of the } x_i \text{ exceed } \sigma X. \end{aligned} \quad (\text{A.68})$$

Indeed, a retained bounded bottom has $x_B \leq X_0 \leq \sigma X$, and a retained narrow bottom that is not covered by the direct estimate below has $x_B < \sigma X$.

There are two direct cases. If $\mathcal{K} \geq X/(2C_1)$, (A.65) gives $\ell_T \geq c_Z X - C_Z$. If a narrow bottom has $x_B \geq \sigma X$, then (A.64) and (A.66) give the same form of estimate, with another constant depending only on Z and σ . After increasing the fixed scale threshold, either estimate implies $\ell_T \geq c_Z H_\alpha(X)$. No estimate on the descendants is used in these two cases.

In every other case, (A.67) gives

$$\sum_i x_i \geq a_0 X. \quad (\text{A.69})$$

Choose $\alpha = \alpha_Z \in (0, 1)$ sufficiently small that, for some fixed $\zeta > 0$,

$$2\sigma^\alpha > 1 + \zeta, \quad \frac{a_0}{\sigma} \sigma^\alpha > 1 + \zeta.$$

This is possible because $a_0/\sigma > 8$ and $\sigma^\alpha \rightarrow 1$ as $\alpha \downarrow 0$. If at least two children exceed σX , monotonicity gives

$$\sum_i H_\alpha(x_i) \geq 2H_\alpha(\sigma X) = 2\sigma^\alpha H_\alpha(X) > (1 + \zeta)H_\alpha(X)$$

once $\sigma X \geq 1$. If all children are at most σX , concavity and $H_\alpha(0) = 0$ give the chord bound

$$H_\alpha(t) \geq \frac{t}{\sigma X} H_\alpha(\sigma X), \quad 0 \leq t \leq \sigma X.$$

Using (A.69),

$$\sum_i H_\alpha(x_i) \geq \frac{a_0}{\sigma} H_\alpha(\sigma X) > (1 + \zeta)H_\alpha(X). \quad (\text{A.70})$$

Thus every case not covered by a direct estimate satisfies a strict H_α -gain.

We finish by strong induction on the number of vertices. The induction claim is that, simultaneously for every rooted tree U , every $0 < z \leq Z$, and every $|\theta| \leq \pi$,

$$\ell_U \geq c_Z H_{\alpha_Z}(x_U),$$

with constants depending only on Z . Choose the global scale threshold large enough to absorb the additive constants in (A.65)–(A.66), to ensure $\sigma X \geq 1$ in the high-scale argument, and to contain X_0/σ . Equation (A.39), with this fixed bounded range, proves the induction claim below the threshold. The two direct cases above prove it at high scale without an induction hypothesis. In every remaining case, all members of $\mathcal{C}(U)$ are pairwise disjoint strict descendant subtrees, so the strong induction hypothesis applies to each. The exact loss comparison above and (A.70) give

$$\ell_U \geq \sum_{W \in \mathcal{C}(U)} \ell_W \geq c_Z \sum_{W \in \mathcal{C}(U)} H_{\alpha_Z}(x_W) \geq c_Z H_{\alpha_Z}(x_U).$$

Taking c_Z to be the minimum of the bounded-scale and direct-estimate constants proves (A.3) for every finite tree. The case $\theta = 0$ has both sides equal to zero.

For a forest with tree components T_i , variances and logarithmic losses add. Concavity and $H_\alpha(0) = 0$ imply subadditivity:

$$H_\alpha\left(\sum_i x_i\right) \leq \sum_i H_\alpha(x_i).$$

Indeed, for $a, b \geq 0$, concavity gives $H_\alpha(a) \geq aH_\alpha(a+b)/(a+b)$ and $H_\alpha(b) \geq bH_\alpha(a+b)/(a+b)$; finite iteration proves the displayed inequality.

Hence the same estimate holds for every finite forest, completing the proof of (A.3). Finally,

$$\inf_{x \geq 0} \frac{H_{\alpha_Z}(x)}{\log(1+x)} > 0,$$

with the ratio at $x = 0$ interpreted by continuity. This proves (A.4) and Theorem A.1.

□

A.7. Aggregate tail consequence

For Corollary A.2, let F_n be the disjoint union of the forests $F_{n,j}$. Its hard-core count is the sum of the independent counts in the corollary, its variance is W_n , and its centered characteristic function is Φ_n . Theorem A.1 therefore gives, for $|u| \leq \pi\sqrt{W_n}$,

$$\left| \Phi_n\left(\frac{u}{\sqrt{W_n}}\right) \right| \leq \exp\{-c_Z \min(u^2, |u|^{2\alpha_Z})\}.$$

The right-hand side is integrable against $1 + u^2$. Its integral over $|u| \geq R$ tends to zero as $R \rightarrow \infty$, uniformly in n , which is (A.5). □

The proof above is self-contained for the hard-core model on finite forests. Its only general-purpose inputs are finite conditional expectation and total variance, Cauchy–Schwarz, Markov’s inequality, the finite Doob maximal inequality proved in Lemma A.3, elementary trigonometric and concavity inequalities, and finite matrix multiplication. In particular, the small-frequency curvature estimate, the uniform characteristic-function gap, and the bounded-scale compactness lemma are all proved here with their full uniform quantifiers. No additional specialized theorem is invoked without a proof in this appendix.

Appendix B. Compact limits and low-activity Gaussianity

This appendix proves the two analytic theorems used in Appendix C. Section B.1 establishes the canonical compact-limit and local-curvature formula. Section B.2 proves a central limit theorem at activity at most $3/2$, with constants independent of the order, component count, degrees, depths, and topology of the forest.

B.1. Canonical compact limits and curvature inversion

For a finite forest F , write

$$I_F(t) = \sum_{k=0}^{\alpha(F)} i_k(F) t^k, \quad (\text{B.1})$$

where $i_k(F)$ is the number of independent k -subsets of $V(F)$ and $\alpha(F)$ is the independence number. At activity $z > 0$, the hard-core law is

$$\Pr_z(I) = \frac{z^{|I|}}{I_F(z)}, \quad I \subseteq V(F) \text{ independent}, \quad (\text{B.2})$$

and $X = |I|$ denotes the total number of occupied vertices. If the coefficient sequence falls and then rises, define

$$m = \min\{k : i_k > i_{k+1}\}, \quad s = \min\{k > m : i_k < i_{k+1}\}.$$

The integer s is the first-recovery index. Its minimality gives

$$i_{s-1} \geq i_s < i_{s+1}, \quad i_s^2 < i_{s-1} i_{s+1}. \quad (\text{B.3})$$

The canonical activity is the unique $z > 0$ for which $\mathbb{E}_z X = s$, and the canonical variance is $V = \text{Var}_z X$. Existence and uniqueness follow from

$$\frac{d}{d \log z} \mathbb{E}_z X = \text{Var}_z X > 0$$

and from the limits of $\mathbb{E}_z X$ as z tends to zero and infinity. Thus a canonical first-recovery sequence (F_n, s_n, z_n, V_n) comes with integer-valued random variables X_n satisfying

$$\mathbb{E} X_n = s_n, \quad \text{Var } X_n = V_n. \quad (\text{B.4})$$

For $j \in \mathbb{Z}$, set

$$p_n(j) = \Pr(X_n = s_n + j), \quad \Delta_n = p_n(0)^2 - p_n(-1)p_n(1). \quad (\text{B.5})$$

The activity factors in (B.2) cancel from the adjacent log-concavity difference:

$$\Delta_n = \frac{z_n^{2s_n}}{I_{F_n}(z_n)^2} (i_{s_n}(F_n)^2 - i_{s_n-1}(F_n) i_{s_n+1}(F_n)) < 0.$$

We use one result from Appendix A and no other Fourier-decay input. Theorem A.1, specialized to the activity ceiling 27, supplies constants $c_{27} > 0$ and $\alpha_{27} \in (0, 1)$ such that, for every finite forest at activity $0 < z < 27$,

$$|\mathbb{E}_z e^{i\theta(X - \mathbb{E}_z X)}| \leq \exp\{-c_{27} \min(V\theta^2, (V\theta^2)^{\alpha_{27}})\}, \quad |\theta| \leq \pi. \quad (\text{B.6})$$

This full fundamental-domain estimate, rather than a compact-frequency version of it, is essential below.

Theorem B.1 (canonical local-limit curvature theorem). Let (F_n, s_n, z_n, V_n) be a canonical first-recovery sequence such that $0 < z_n < 27$ and $V_n \rightarrow \infty$. From every subsequence one can extract a further subsequence, indexed again by n , for which

$$Y_n := \frac{X_n - s_n}{\sqrt{V_n}} \Rightarrow Y \quad (\text{B.7})$$

for a probability law having a C^2 density f . If

$$\phi_n(u) = \mathbb{E} e^{iuY_n}, \quad D_n = [-\pi\sqrt{V_n}, \pi\sqrt{V_n}], \quad \tilde{\phi}_n = \mathbf{1}_{D_n} \phi_n, \quad (\text{B.8})$$

and ϕ is the characteristic function of Y , then, for $m \in \{0, 1, 2\}$,

$$\int_{\mathbb{R}} (1 + |u|^m) |\tilde{\phi}_n(u) - \phi(u)| du \rightarrow 0. \quad (\text{B.9})$$

Moreover,

$$V_n^2 [p_n(0)^2 - p_n(-1)p_n(1)] \rightarrow (f'(0))^2 - f(0)f''(0). \quad (\text{B.10})$$

When $f(0) > 0$, the expression on the right is $f(0)^2 [-(\log f)''(0)]$. This identity explains the use of the term curvature in the theorem title; no separate notion of curvature is being introduced.

In particular, every such limiting density satisfies $(f'(0))^2 - f(0)f''(0) \leq 0$.

Proof. Begin with an arbitrary subsequence. By (B.4), $\mathbb{E}Y_n = 0$ and $\mathbb{E}Y_n^2 = 1$. Chebyshev's inequality makes the laws of Y_n tight, so Prokhorov's theorem gives a further weakly convergent subsequence. Denote its limiting law by that of Y . The uniform second-moment bound also makes (Y_n) uniformly integrable in L^1 , so $\mathbb{E}Y = 0$; lower semicontinuity gives $\mathbb{E}Y^2 \leq 1$. No equality of limiting variances is needed.

Define

$$H_{27}(u) = \exp\{-c_{27} \min(u^2, |u|^{2\alpha_{27}})\}. \quad (\text{B.11})$$

Substitution of $\theta = u/\sqrt{V_n}$ into (B.6), using the exact centering in (B.4), gives

$$|\phi_n(u)| \leq H_{27}(u), \quad u \in D_n. \quad (\text{B.12})$$

The weights 1, $|u|$, and $|u|^2$ are integrable against H_{27} . Weak convergence and the Lévy continuity theorem give $\phi_n(u) \rightarrow \phi(u)$ for each fixed u . Since the intervals D_n exhaust $\mathbb{R}$, (B.12) also gives $|\phi(u)| \leq H_{27}(u)$. Therefore

$$(1 + |u|^m)|\tilde{\phi}_n(u) - \phi(u)| \leq 2(1 + |u|^m)H_{27}(u), \quad (\text{B.13})$$

and dominated convergence proves (B.9).

In particular, $|u|^m \phi(u) \in L^1(\mathbb{R})$ for $m = 0, 1, 2$. Fourier inversion identifies the limiting law with the density

$$f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-iux} \phi(u) du, \quad f^{(m)}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} (-iu)^m e^{-iux} \phi(u) du. \quad (\text{B.14})$$

Thus $f \in C^2(\mathbb{R})$.

It remains to prove the local statement. Because s_n is an integer, lattice Fourier inversion on the scaled fundamental domain gives, for every $j \in \mathbb{Z}$,

$$p_n(j) = \frac{1}{2\pi\sqrt{V_n}} \int_{D_n} e^{-iuj/\sqrt{V_n}} \phi_n(u) du. \quad (\text{B.15})$$

Applying (B.15) to $j = -1, 0, 1$, taking the real part, and using the zero extension in (B.8) yields the exact identity

$$V_n^2 \Delta_n = \frac{1}{(2\pi)^2} \operatorname{Re} \iint_{\mathbb{R}^2} K_{V_n}(u, v) \tilde{\phi}_n(u) \tilde{\phi}_n(v) du dv, \quad (\text{B.16})$$

where

$$K_V(u, v) = V \left(1 - \cos \frac{u-v}{\sqrt{V}} \right). \quad (\text{B.17})$$

Indeed, the imaginary sine part of $V_n[1 - e^{i(u-v)/\sqrt{V_n}}]$ integrates to zero exactly: $\phi_n(u)\phi_n(v)$ and $D_n \times D_n$ are invariant under interchanging u and v , whereas $\sin((u-v)/\sqrt{V_n})$ is antisymmetric.

There is no complex conjugate in (B.16): the product $p_n(-1)p_n(1)$ produces the phase $e^{i(u-v)/\sqrt{V_n}}$. For all u, v ,

$$0 \leq K_{V_n}(u, v) \leq \frac{(u-v)^2}{2}, \quad K_{V_n}(u, v) \rightarrow \frac{(u-v)^2}{2}. \quad (\text{B.18})$$

The absolute value of the integrand in (B.16) is consequently bounded by

$$\frac{(u-v)^2}{2} H_{27}(u) H_{27}(v), \quad (\text{B.19})$$

which is integrable on $\mathbb{R}^2$. Dominated convergence, equivalently the weighted convergence (B.9) with $m = 2$, gives

$$\lim_{n \rightarrow \infty} V_n^2 \Delta_n = \frac{1}{(2\pi)^2} \operatorname{Re} \iint_{\mathbb{R}^2} \frac{(u-v)^2}{2} \phi(u) \phi(v) du dv. \quad (\text{B.20})$$

From (B.14),

$$f'(0) = \frac{1}{2\pi} \int_{\mathbb{R}} (-iu) \phi(u) du, \quad f''(0) = \frac{1}{2\pi} \int_{\mathbb{R}} (-u^2) \phi(u) du. \quad (\text{B.21})$$

Expanding $(u-v)^2/2$ in (B.20) and symmetrizing shows that its right-hand side is exactly $(f'(0))^2 - f(0)f''(0)$. This proves (B.10). Finally, $\Delta_n < 0$ for every n , so the limiting curvature is nonpositive. $\square$

The centering in Theorem B.1 is exact: the canonical activity is chosen so that the integer s_n equals $\mathbb{E}X_n$. More generally, if an integer center k_n is used and $(X_n - k_n)/\sqrt{V_n}$ is tight, then

$$m_n = \frac{\mathbb{E}X_n - k_n}{\sqrt{V_n}} \quad (\text{B.22})$$

is bounded. Indeed, for $R < |m_n|$, Chebyshev's inequality gives

$$\Pr\left(\left|\frac{X_n - k_n}{\sqrt{V_n}}\right| \leq R\right) \leq (|m_n| - R)^{-2}. \quad (\text{B.23})$$

After extraction $m_n \rightarrow m$, and Slutsky's theorem translates any weak limit centered at the exact mean by m . In the canonical normalization above, $m_n = 0$ identically.

B.2. A uniform central limit theorem at low activity

Theorem B.2 (low-activity CLT). Let F_n be arbitrary finite forests, let $0 < z_n \leq 3/2$, and let X_n be the total occupation count under the hard-core law on F_n at activity z_n . If

$$V_n := \text{Var } X_n \rightarrow \infty,$$

then

$$\frac{X_n - \mathbb{E}X_n}{\sqrt{V_n}} \Rightarrow N(0, 1). \quad (\text{B.24})$$

Proof. Fix a finite forest F , root every component arbitrarily, and give each component root a fictitious absent parent. Order the components arbitrarily and, within each component, order every parent before its children. We establish estimates for this rooted forest at a fixed activity $0 < z \leq 3/2$. All constants below are absolute and independent of F , its number of components, and z in this range.

Boundary-conditioned recursion and martingale representation. For a vertex u , let T_u be its descendant subtree. Write P_u for the partition function of the unrestricted hard-core law on T_u when the parent of u is absent, Q_u for the partition function with u forced absent, and R_u for the partition function with u forced occupied. If c ranges over the children of u , then

$$P_u = Q_u + R_u, \quad Q_u = \prod_c P_c, \quad R_u = z \prod_c Q_c. \quad (\text{B.25})$$

Set

$$p_u = \frac{R_u}{P_u}, \quad q_u = 1 - p_u = \frac{Q_u}{P_u}, \quad \ell_u = \frac{p_u}{q_u}.$$

The exact rooted-tree recursion for the occupation odds is

$$\ell_u = z \prod_c q_c, \quad 0 < \ell_u \leq z. \quad (\text{B.26})$$

Let $\mu_u^P, \mu_u^Q, \mu_u^R$ denote the expected occupation count in T_u in the three states just defined, and put

$$\delta_u = \mu_u^R - \mu_u^Q. \quad (\text{B.27})$$

Since $\mu_u^Q = \sum_c \mu_c^P$, $\mu_u^R = 1 + \sum_c \mu_c^Q$, and $\mu_c^P = q_c \mu_c^Q + p_c \mu_c^R$, these conditional mean differences obey

$$\delta_u = 1 - \sum_c p_c \delta_c. \quad (\text{B.28})$$

Let ξ_u be the occupation indicator of u . Define

$$a_u = \Pr(\text{the parent of } u \text{ is absent}), \quad w_u = a_u p_u q_u, \quad (\text{B.29})$$

with $a_r = 1$ at every component root. For every child c of u ,

$$a_c = \Pr(\xi_u = 0) = 1 - a_u p_u. \quad (\text{B.30})$$

Order the vertices so that every parent precedes its children, and let $\mathcal{F}_{u-}$ denote the sigma-field generated by the occupation variables revealed before u . The forest Markov property, component independence, and (B.25) give

$$\mathbb{E}(\xi_u \mid \mathcal{F}_{u-}) = p_u (1 - \xi_{\text{par}(u)}).$$

Consequently

$$\eta_u = \xi_u - p_u (1 - \xi_{\text{par}(u)}) \quad (\text{B.31})$$

is a martingale difference and

$$\mathbb{E}(\eta_u^2 \mid \mathcal{F}_{u-}) = p_u q_u (1 - \xi_{\text{par}(u)}), \quad \mathbb{E} \eta_u^2 = w_u. \quad (\text{B.32})$$

In $\sum_u \delta_u \eta_u$, the coefficient of ξ_u is $\delta_u + \sum_c p_c \delta_c = 1$ by (B.28). The sum has mean zero, so its constant term is $-\mathbb{E}X$. Orthogonality of martingale differences therefore gives the exact identities

$$X - \mathbb{E}X = \sum_u \delta_u \eta_u, \quad V := \text{Var } X = \sum_u w_u \delta_u^2. \quad (\text{B.33})$$

Weighted transfer contractions. We next prove two contraction estimates that are uniform over the forest. For a vector $x = (x_u)_{u \in F}$, define the upward transfer operator and weighted norms by

$$(\mathcal{T}x)_u = \sum_{c \text{ child of } u} p_c x_c, \quad \|x\|_{r,w}^r = \sum_u w_u |x_u|^r. \quad (\text{B.34})$$

For all children c of a fixed u , (B.30) gives

$$w_c = A_u p_c q_c, \quad A_u := 1 - a_u p_u, \quad \frac{w_u}{A_u} \leq p_u, \quad (\text{B.35})$$

because $a_u q_u \leq 1 - a_u p_u$. Cauchy–Schwarz now gives

$$\left| \sum_c p_c x_c \right|^2 \leq \left(\sum_c \frac{p_c}{q_c} \right) \left(\sum_c p_c q_c x_c^2 \right), \quad (\text{B.36})$$

and hence

$$w_u |(\mathcal{T}x)_u|^2 \leq K_2(u) \sum_c w_c x_c^2, \quad K_2(u) = p_u \sum_c \ell_c. \quad (\text{B.37})$$

Put

$$Y_u = \sum_c \log(1 + \ell_c).$$

Since $q_c = (1 + \ell_c)^{-1}$, (B.26) implies

$$p_u = \frac{z}{z + e^{Y_u}}. \quad (\text{B.38})$$

The function $r / \log(1 + r)$ is increasing on $(0, \infty)$. Since $\ell_c \leq z$,

$$\sum_c \ell_c \leq \frac{z}{\log(1 + z)} Y_u.$$

Together with $Y_u e^{-Y_u} \leq e^{-1}$, this gives

$$K_2(u) \leq \frac{z^2}{e \log(1 + z)} \leq \frac{9}{4e \log(5/2)} < \frac{15}{16}. \quad (\text{B.39})$$

The middle expression is increasing in z . The last strict inequality follows, for example, from $e > 8/3$ and $\log(5/2) > 9/10$. Summing (B.37) over parents, each nonroot child is counted once, and therefore

$$\|\mathcal{T}x\|_{2,w}^2 \leq \frac{15}{16} \|x\|_{2,w}^2. \quad (\text{B.40})$$

For the cubic estimate, Hölder's inequality gives

$$\left| \sum_c p_c x_c \right|^3 \leq \left(\sum_c p_c q_c |x_c|^3 \right) \left(\sum_c \frac{p_c}{\sqrt{q_c}} \right)^2. \quad (\text{B.41})$$

Using (B.35),

$$w_u |(\mathcal{T}x)_u|^3 \leq K_3(u) \sum_c w_c |x_c|^3,$$

where

$$K_3(u) = p_u \left(\sum_c \frac{p_c}{\sqrt{q_c}} \right)^2 = p_u \left(\sum_c \frac{\ell_c}{\sqrt{1 + \ell_c}} \right)^2. \quad (\text{B.42})$$

Writing $y = \log(1 + r)$, one has $r/\sqrt{1 + r} = 2 \sinh(y/2)$, and $2 \sinh(y/2)/y$ is increasing for $y > 0$. Hence

$$\sum_c \frac{\ell_c}{\sqrt{1 + \ell_c}} \leq \frac{z}{\sqrt{1 + z} \log(1 + z)} Y_u.$$

Equations (B.38) and $Y_u^2 e^{-Y_u} \leq 4e^{-2}$ yield

$$K_3(u) \leq \frac{4z^3}{e^2(1 + z) \log^2(1 + z)} \leq \frac{27}{5e^2 \log^2(5/2)} < \frac{15}{16}. \quad (\text{B.43})$$

The first expression is increasing in z : its logarithmic derivative has the sign of $(3 + 2z) \log(1 + z) - 2z > 0$. The last inequality follows from $e \log(5/2) > 12/5$, as in (B.39). Summing over parents gives

$$\|\mathcal{T}x\|_{3,w}^3 \leq \frac{15}{16} \|x\|_{3,w}^3. \quad (\text{B.44})$$

Moments of the conditional mean differences. The gap recursion (B.28) is

$$\mathbf{1} = (I + \mathcal{T})\delta. \quad (\text{B.45})$$

Let

$$W = \sum_u w_u, \quad S_3 = \sum_u w_u |\delta_u|^3.$$

The triangle inequality, (B.33), and (B.40) imply

$$W^{1/2} = \|\mathbf{1}\|_{2,w} \leq \left(1 + \frac{\sqrt{15}}{4} \right) \|\delta\|_{2,w} = \left(1 + \frac{\sqrt{15}}{4} \right) V^{1/2}.$$

Similarly, (B.44)–(B.45) give

$$S_3^{1/3} \leq W^{1/3} + \left(\frac{15}{16} \right)^{1/3} S_3^{1/3}.$$

Thus

$$W \leq C_W V, \quad S_3 \leq C_3 V, \quad (\text{B.46})$$

where

$$C_W = \left(1 + \frac{\sqrt{15}}{4}\right)^2, \quad C_3 = \frac{C_W}{(1 - (15/16)^{1/3})^3}. \quad (\text{B.47})$$

Conditional on an absent parent, η_u is a centered Bernoulli variable, and therefore

$$\mathbb{E}(|\eta_u|^3 \mid \xi_{\text{par}(u)} = 0) = p_u q_u (p_u^2 + q_u^2) \leq p_u q_u.$$

If the parent is occupied, $\eta_u = 0$. Consequently

$$\sum_u \mathbb{E}|\delta_u \eta_u|^3 \leq \sum_u w_u |\delta_u|^3 = S_3 \leq C_3 V. \quad (\text{B.48})$$

Predictable quadratic variation. We now concentrate the predictable quadratic variation. From (B.32), it is

$$\mathcal{V}_F = \sum_u p_u q_u \delta_u^2 (1 - \xi_{\text{par}(u)}), \quad \mathbb{E}\mathcal{V}_F = V, \quad (\text{B.49})$$

where every component-root parent indicator is fixed at zero. Set

$$h_u = p_u q_u \delta_u^2, \quad A_x = \sum_{c \text{ child of } x} h_c.$$

The component-root terms are deterministic, and reindexing all other parent indicators gives

$$\mathcal{V}_F = \text{a deterministic constant} - \sum_x A_x \xi_x. \quad (\text{B.50})$$

We need a variance bound for deterministic linear scores. Given coefficients d_u , let $Y_d = \sum_u d_u \xi_u$, and define θ recursively by

$$\theta_u = d_u - \sum_c p_c \theta_c.$$

The coefficient calculation used in (B.33) gives

$$Y_d - \mathbb{E}Y_d = \sum_u \theta_u \eta_u, \quad \text{Var } Y_d = \sum_u w_u \theta_u^2. \quad (\text{B.51})$$

In vector notation, $\theta = (I + \mathcal{T})^{-1}d$. The operator $\mathcal{T}$ is the direct sum of the component transfer operators and is nilpotent on a finite rooted forest. Thus (B.40) gives

$$\text{Var } Y_d \leq \frac{1}{(1 - \sqrt{15}/4)^2} \sum_u w_u d_u^2. \quad (\text{B.52})$$

Applying (B.52) to $d_x = A_x$ and using (B.50),

$$\text{Var } \mathcal{V}_F \leq \frac{1}{(1 - \sqrt{15}/4)^2} \sum_x w_x A_x^2. \quad (\text{B.53})$$

For each x , define

$$S_x = \sum_c p_c q_c, \quad B_x = \sum_c p_c q_c |\delta_c|^3.$$

Hölder interpolation gives

$$A_x \leq S_x^{1/3} B_x^{2/3}. \quad (\text{B.54})$$

If $t_x = \sum_c p_c$, then $S_x \leq t_x$, while (B.26) and $q_c \leq e^{-p_c}$ imply

$$w_x \leq p_x \leq z \prod_c q_c \leq z e^{-t_x}.$$

Since $\max_{t \geq 0} e^{-t} t^{2/3} = (2/(3e))^{2/3}$,

$$w_x S_x^{2/3} \leq \frac{3}{2} \left(\frac{2}{3e} \right)^{2/3} =: C_B. \quad (\text{B.55})$$

For each child c of x ,

$$a_c = 1 - a_x p_x \geq q_x = \frac{1}{1 + \ell_x} \geq \frac{1}{1 + z} \geq \frac{2}{5}.$$

Hence

$$B_x \leq \frac{5}{2} \sum_{c \text{ child of } x} w_c |\delta_c|^3, \quad \sum_x B_x \leq \frac{5}{2} S_3 \leq \frac{5}{2} C_3 V. \quad (\text{B.56})$$

The child sets of distinct parents are disjoint. Using (B.54)–(B.56) and $\sum_x B_x^{4/3} \leq (\sum_x B_x)^{4/3}$, we obtain

$$\sum_x w_x A_x^2 \leq C_B \sum_x B_x^{4/3} \leq C_B \left(\frac{5}{2} C_3 V \right)^{4/3}. \quad (\text{B.57})$$

Equations (B.49), (B.53), and (B.57) show that an absolute constant C_Q satisfies

$$\text{Var } \mathcal{V}_F \leq C_Q V^{4/3}, \quad \mathbb{E} \left| \frac{\mathcal{V}_F}{V} - 1 \right|^2 \leq C_Q V^{-2/3}. \quad (\text{B.58})$$

Martingale limit. Return now to the sequence (F_n, z_n) . In each row, concatenate the componentwise parent-before-child orders and define

$$D_{n,u} = \frac{\delta_{n,u} \eta_{n,u}}{\sqrt{V_n}}.$$

By (B.33), their row sum is $(X_n - \mathbb{E}X_n)/\sqrt{V_n}$. By (B.58), their predictable quadratic variation satisfies

$$\sum_u \mathbb{E}(D_{n,u}^2 \mid \mathcal{F}_{n,u^-}) = \frac{\mathcal{V}_{F_n}}{V_n} \longrightarrow 1 \quad \text{in } L^2. \quad (\text{B.59})$$

For every $\varepsilon > 0$, the conditional Lindeberg sum $L_n(\varepsilon)$ obeys

$$\begin{aligned} L_n(\varepsilon) &:= \sum_u \mathbb{E}\left(D_{n,u}^2 \mathbf{1}_{\{|D_{n,u}| > \varepsilon\}} \mid \mathcal{F}_{n,u^-}\right), \\ \mathbb{E}L_n(\varepsilon) &\leq \frac{1}{\varepsilon} \sum_u \mathbb{E}|D_{n,u}|^3 \leq \frac{C_3}{\varepsilon \sqrt{V_n}} \longrightarrow 0. \end{aligned} \quad (\text{B.60})$$

Thus $L_n(\varepsilon) \rightarrow 0$ in probability. We use the following standard constant-variance form of the martingale triangular-array central limit theorem: if each row is a finite square-integrable martingale-difference array, its predictable quadratic variation converges in probability to one, and its conditional Lindeberg sums converge in probability to zero, then its row sum converges weakly to $N(0, 1)$. This is [HH80, Corollary 3.1, p. 58, with the subsequent remark on p. 59 that removes the cross-row nesting condition (3.21) when the limiting variance is a nonrandom constant]. Equation (B.59) is exactly the predictable-variance hypothesis and (B.60) is the conditional Lindeberg hypothesis. The theorem gives (B.24), completing the proof. $\square$

B.3. Classical inputs

The compactness, convergence, and Fourier steps use Prokhorov's theorem, the Lévy continuity theorem, Slutsky's theorem, and Fourier inversion for integrable characteristic functions and integer-valued laws; see [Bil99] and [Fel71]. The precise constant-variance form of the martingale triangular-array central limit theorem used after (B.59)–(B.60) is stated there and attributed to [HH80, Corollary 3.1, p. 58, and the subsequent remark on p. 59].

Appendix C. Occupation balance and the scale of first recovery

This appendix proves two quantitative inputs used after Theorem 4.1. The first is a uniform occupation-probability bound at a vertex with a dominant variance contribution. The second compares the size, mean, and variance of a canonical first-recovery state. All conditional laws below are taken at the activity of the original global forest. In particular, no component, descendant subtree, or conditional binomial law is assigned a first-recovery property.

C.1. Canonical states and two analytic theorems

Let F be a finite forest of order N , with independence polynomial

$$I_F(t) = \sum_{k=0}^{\alpha(F)} i_k(F) t^k,$$

where $i_k(F)$ is the number of independent k -subsets of $V(F)$ and $\alpha(F)$ is the independence number. If the coefficient sequence falls and then rises, put

$$m = \min\{k : i_k > i_{k+1}\}, \quad s = \min\{k > m : i_k < i_{k+1}\}.$$

Thus

$$i_{s-1} \geq i_s < i_{s+1}, \quad i_s^2 < i_{s-1} i_{s+1}. \quad (\text{C.1})$$

For $z > 0$, the hard-core law on F is

$$\Pr_z(I) = \frac{z^{|I|}}{I_F(z)}, \quad I \subseteq V(F) \text{ independent}.$$

Write $X = |I|$, and let $\xi_u = \mathbf{1}_{\{u \in I\}}$ be the occupation indicator of u . Since

$$\frac{d}{d \log z} \mathbb{E}_z X = \text{Var}_z X > 0,$$

there is a unique activity z for which $\mathbb{E}_z X = s$. We write

$$V = \text{Var}_z X, \quad \pi(j) = \Pr_z(X = s + j).$$

The quadruple (F, s, z, V) is the **canonical first-recovery state**. By (C.1), its central log-concavity difference satisfies

$$\mathcal{D}_s := \pi(0)^2 - \pi(-1)\pi(1) < 0. \quad (\text{C.2})$$

A **canonical first-recovery sequence** is any sequence of these states. As in Section 1, it is called **variance-divergent** when $V_n \rightarrow \infty$; activity restrictions are always stated separately.

We use the following two theorems proved in Appendix B. Theorem B.1 says that from every subsequence of a variance-divergent canonical sequence with $0 < z_n < 27$, one may

extract a further subsequence for which $(X_n - s_n)/\sqrt{V_n}$ converges in distribution to a law with a C^2 density f , and

$$V_n^2 \mathcal{D}_{s_n} \longrightarrow (f'(0))^2 - f(0)f''(0). \quad (\text{C.3})$$

Theorem B.2 says that for arbitrary finite forests and activities $0 < z_n \leq 3/2$, if $V_n \rightarrow \infty$, then

$$\frac{X_n - \mathbb{E}X_n}{\sqrt{V_n}} \Rightarrow N(0, 1). \quad (\text{C.4})$$

Lemma C.1 depends on both proved theorems: Theorem B.2 identifies the only possible low-activity limit, while Theorem B.1 transfers the strict first-recovery log-concavity difference to its limiting curvature.

Lemma C.1 (activity lower bound at large variance). There is $V_{\text{act}} < \infty$ such that every canonical first-recovery state with $0 < z < 27$ and $V \geq V_{\text{act}}$ satisfies

$$z > \frac{3}{2}. \quad (\text{C.5})$$

Proof. Otherwise there would be a variance-divergent canonical sequence with $z_n \leq 3/2$. Since $\mathbb{E}X_n = s_n$, (C.4) makes $(X_n - s_n)/\sqrt{V_n}$ converge to the standard normal law. Apply (C.3) on a further subsequence. Its limiting density is $\varphi(x) = (2\pi)^{-1/2}e^{-x^2/2}$, and hence

$$(\varphi'(0))^2 - \varphi(0)\varphi''(0) = \varphi(0)^2 = \frac{1}{2\pi} > 0.$$

This contradicts (C.2), because every term $V_n^2 \mathcal{D}_{s_n}$ is negative. The sequential contradiction also gives the uniform threshold V_{act} . $\square$

C.2. Boundary-conditioned laws and recursive variance mass

In Sections C.2–C.3 the forest need not be canonical; V denotes the variance of its global hard-core count at the displayed activity.

Root every component of F , and write $\text{par}(u)$ for the parent of a nonroot vertex u . A vertex together with this rooted inherited context will be understood throughout. Let T_u be the descendant subtree rooted at u . Define the three boundary-conditioned partition functions

$$P_u(t) = I_{T_u}(t), \quad Q_u(t) = \prod_{v \text{ child of } u} P_v(t), \quad R_u(t) = t \prod_{v \text{ child of } u} Q_v(t).$$

Thus $P_u = Q_u + R_u$. The P_u -state is the unrestricted hard-core law on T_u when the parent of u is absent; the Q_u -state forces u absent; and the R_u -state forces u occupied. At the fixed activity z , set

$$p_u = \frac{R_u(z)}{P_u(z)}, \quad q_u = 1 - p_u.$$

The rooted recursion for the conditional occupation probability is

$$\frac{p_u}{q_u} = z \prod_{v \text{ child of } u} q_v. \quad (\text{C.6})$$

For $0 < z < 27$, it implies

$$q_u > \frac{1}{28}, \quad p_u < \frac{27}{28}. \quad (\text{C.7})$$

Let $\mu_u^P, \mu_u^Q, \mu_u^R$ and v_u^P, v_u^Q, v_u^R denote the count means and variances in these three states, and define the occupied-versus-absent displacement

$$\Delta_u = \mu_u^R - \mu_u^Q. \quad (\text{C.8})$$

Let a_u be the probability, in the original globally rooted law, that the parent of u is absent, with $a_r = 1$ at every component root, and put $b_u = 1 - a_u$. If v is a child of u , then

$$a_v = a_u q_u + b_u = 1 - a_u p_u, \quad b_v = a_u p_u. \quad (\text{C.9})$$

In particular, every nonroot vertex satisfies

$$a_u > \frac{1}{28}. \quad (\text{C.10})$$

Define the **subtree variance mass** and the **vertex variance contribution** at u by

$$E(u) = a_u v_u^P + b_u v_u^Q, \quad g(u) = a_u p_u q_u \Delta_u^2. \quad (\text{C.11})$$

These definitions retain the context in which T_u occurs inside the original forest. The conditional-variance identity and the product structure of the child subtrees give

$$v_u^P = q_u v_u^Q + p_u v_u^R + p_u q_u \Delta_u^2,$$

$$v_u^Q = \sum_v v_v^P, \quad v_u^R = \sum_v v_v^Q.$$

Substituting (C.9) yields the recursive variance decomposition

$$E(u) = g(u) + \sum_{v \text{ child of } u} E(v), \quad \sum_{r \in \mathcal{R}(F)} E(r) = V, \quad (\text{C.12})$$

where $\mathcal{R}(F)$ is the set of component roots. Consequently

$$g(u) \leq E(u) \leq V. \quad (\text{C.13})$$

Moreover, (C.10) gives the conditional-variance estimate

$$v_u^P \leq 28E(u) \quad (\text{C.14})$$

for every nonroot u , while at a component root it holds because $E(r) = v_r^P$.

C.3. Large conditional mean differences

We next prove the local estimate that prevents a dominant vertex variance contribution from being carried by a vanishing occupation probability. Retain the notation $a_u, p_u, q_u, \Delta_u, g(u)$, and V from Section C.2. The statement applies to a rooted forest under its unrestricted hard-core law, with a fictitious forced-absent parent at each component root. The conditional means satisfy

$$\Delta_u = 1 - \sum_{v \text{ child of } u} p_v \Delta_v. \quad (\text{C.15})$$

If vertices are ordered with parents before children and

$$\eta_u = \xi_u - p_u(1 - \xi_{\text{par}(u)}),$$

then (η_u) are martingale differences. Equation (C.15) collects the coefficient of every occupation indicator in the exact decomposition

$$X - \mathbb{E}X = \sum_u \Delta_u \eta_u.$$

Since $\mathbb{E}\eta_u^2 = a_u p_u q_u$, orthogonality gives

$$V = \text{Var } X = \sum_u g(u). \quad (\text{C.16})$$

For $b > 1$, define the contribution from large conditional mean differences

$$G(b) = \sum_{u: |\Delta_u| > b} g(u). \quad (\text{C.17})$$

Lemma C.2 (large-displacement contribution bound). Put

$$\chi = \frac{784V}{b^2}.$$

If $b > 1$ and $\chi < 1$, then

$$\frac{G(b)}{V} \leq \left(\frac{b}{b-1} \right)^2 28\chi \log \frac{27}{\chi}. \quad (\text{C.18})$$

In particular, uniformly over all finite forest topologies and all $0 < z < 27$,

$$b \rightarrow \infty, \quad \frac{V}{b^2} \rightarrow 0 \quad \Rightarrow \quad \frac{G(b)}{V} \rightarrow 0. \quad (\text{C.19})$$

Proof. For arbitrary real y_v on the children of u , weighted Cauchy-Schwarz gives

$$\left(\sum_v p_v y_v \right)^2 \leq \left(\sum_v \frac{p_v}{q_v} \right) \left(\sum_v p_v q_v y_v^2 \right). \quad (\text{C.20})$$

All children have parent-absence probability $a_v = 1 - a_u p_u$. Since

$$\frac{a_u p_u q_u}{a_v} = \frac{a_u p_u q_u}{1 - a_u p_u} \leq p_u,$$

(C.20) implies

$$a_u p_u q_u \left(\sum_v p_v y_v \right)^2 \leq p_u \left(\sum_v \frac{p_v}{q_v} \right) \sum_v a_v p_v q_v y_v^2. \quad (\text{C.21})$$

By (C.6), (C.7), and $t \leq -\log(1-t)$ for $0 \leq t < 1$,

$$\sum_v \frac{p_v}{q_v} \leq 28 \sum_v p_v \leq 28 \sum_v (-\log q_v) = 28 \log \frac{z q_u}{p_u} \leq 28 \log \frac{27}{p_u}. \quad (\text{C.22})$$

If $|\Delta_u| > b$, then (C.15) gives

$$\left| \sum_v p_v \Delta_v \right| = |1 - \Delta_u| \geq \left(1 - \frac{1}{b} \right) |\Delta_u|. \quad (\text{C.23})$$

Also $a_u, q_u > 1/28$, with $a_u = 1$ at a component root, so (C.16) implies

$$\frac{p_u \Delta_u^2}{784} \leq g(u) \leq V, \quad p_u < \frac{784V}{b^2} = \chi. \quad (\text{C.24})$$

The function $t \mapsto t \log(27/t)$ is increasing on $(0, 1)$. Applying (C.21)–(C.24) with $y_v = \Delta_v$ therefore gives, for every vertex counted by $G(b)$,

$$g(u) \leq \left(\frac{b}{b-1} \right)^2 28\chi \log \frac{27}{\chi} \sum_{v \text{ child of } u} g(v). \quad (\text{C.25})$$

After summation, every child term on the right occurs at most once, because a vertex has a unique parent. The double sum is at most V by (C.16), which proves (C.18). The right side of (C.18) tends to zero under the hypotheses of (C.19). $\square$

C.4. Occupation and displacement bounds at a dominant vertex

Theorem C.3 (bounds at a dominant vertex). Fix $c > 0$. There are $\rho(c) > 0$ and $V_{\text{bal}}(c) < \infty$ such that the following holds. In any canonical first-recovery state at activity $0 < z < 27$, after rooting each component arbitrarily, if a vertex u satisfies

$$V \geq V_{\text{bal}}(c), \quad g(u) \geq cV, \quad (\text{C.26})$$

then

$$p_u \geq \rho(c), \quad (\text{C.27})$$

and

$$4cV \leq \Delta_u^2 \leq \frac{784}{\rho(c)} V. \quad (\text{C.28})$$

Together with $q_u > 1/28$, (C.27) says that the two local boundary states are uniformly balanced.

Proof. Suppose that no uniform lower bound in (C.27) exists. By contradiction along a sequence, there are canonical first-recovery states, chosen component rootings, and vertices u_n with

$$V_n \rightarrow \infty, \quad g(u_n) \geq cV_n, \quad p_{u_n} \rightarrow 0. \quad (\text{C.29})$$

Apply Lemma C.2 to the local P_{u_n} -state, without changing the activity or giving that state any global recovery property. Write $v = v_{u_n}^P$ and $\Delta = \Delta_{u_n}$. Its root variance contribution is

$$\tilde{g} = p_{u_n} q_{u_n} \Delta^2 = \frac{g(u_n)}{a_{u_n}} \geq cV_n, \quad (\text{C.30})$$

while (C.13)–(C.14) give

$$v \leq 28E(u_n) \leq 28V_n. \quad (\text{C.31})$$

Set $b = |\Delta|/2$. From (C.30) and $p_{u_n} q_{u_n} \leq 1/4$, $b \rightarrow \infty$. Moreover,

$$\frac{784v}{b^2} = \frac{3136v}{\Delta^2} \leq \frac{87808V_n}{\Delta^2} \leq \frac{87808}{c} p_{u_n} q_{u_n} \rightarrow 0. \quad (\text{C.32})$$

Thus Lemma C.2 applies for all sufficiently large n and gives $G(b)/v \rightarrow 0$. The local root itself has gap $|\Delta| > b$, however, so

$$\frac{G(b)}{v} \geq \frac{\tilde{g}}{v} \geq \frac{c}{28}, \quad (\text{C.33})$$

a contradiction. This proves a uniform $\rho(c) > 0$; otherwise choosing $V_n \geq n$ and $p_{u_n} < 1/n$ would reproduce (C.29).

Finally, $a_u p_u q_u \leq 1/4$, so (C.26) gives

$$\Delta_u^2 \geq 4cV.$$

On the other hand, (C.7), (C.10), and (C.27) imply $a_u p_u q_u \geq \rho(c)/784$, with a component-root case even stronger. Since $g(u) \leq E(u) \leq V$,

$$\Delta_u^2 = \frac{g(u)}{a_u p_u q_u} \leq \frac{784}{\rho(c)} V.$$

This proves (C.28). $\square$

In particular, if Theorem 4.1 supplies u with $g(u) \geq \kappa_{27}V$, Theorem C.3 applies with $c = \kappa_{27}$. The constant $\kappa_{27} > 0$ is the one in Theorem 4.1, so the resulting constants depend only on the fixed activity cap.

C.5. Addable vertices, low outdegree, and bipartition conditioning

We now turn to a canonical first-recovery state with $0 < z < 27$ and $V \geq V_{\text{act}}$. By Lemma C.1, throughout this section we may assume

$$\frac{3}{2} < z < 27. \quad (\text{C.34})$$

For an independent set I , a vertex is **addable** if it does not belong to I and has no neighbor in I . Let

$$A = A(I) = \#\{v \notin I : N(v) \cap I = \emptyset\}.$$

For a uniformly chosen independent k -set, write A_k for its number of addable vertices and $\mu_k = \mathbb{E}A_k$. Counting pairs (I, v) , where I has size k and v is addable to I , gives the exact insertion identity

$$i_k \mu_k = (k+1) i_{k+1}. \quad (\text{C.35})$$

Averaging (C.35) under the hard-core law gives the same-activity identity

$$z \mathbb{E}A = \mathbb{E}X = s. \quad (\text{C.36})$$

At the first-recovery index, (C.1) and (C.35) give the stronger conditional expectation identity

$$\mathbb{E}[A \mid X = s] = (s+1) \frac{i_{s+1}}{i_s} > s+1. \quad (\text{C.37})$$

We next record the order bound that is needed both for Hoeffding's inequality and for the upper variance estimate.

Lemma C.4 (low-outdegree order bound). Under (C.34),

$$N+1 < C_* s, \quad C_* := \frac{3304}{3}. \quad (\text{C.38})$$

Proof. Let $q \geq 1$ be the number of components. A componentwise rooted forest has $N - q$ child edges, so the number of vertices with at least two children is at most $(N - q)/2$. Hence at least $(N + q)/2 \geq (N + 1)/2$ vertices have at most one child.

For a vertex with at most one child, (C.6), (C.7), and $z > 3/2$ imply

$$\frac{p_u}{q_u} > \frac{3}{56}, \quad p_u > \frac{3}{59}.$$

The probability that its parent is absent is greater than $1/28$, except at a component root where it equals one. Hence its global occupation probability is greater than $3/1652$. Summing occupation probabilities over the $(N + 1)/2$ such vertices gives

$$s = \mathbb{E}X > \frac{3(N+1)}{3304},$$

which is (C.38). $\square$

Two immediate consequences are

$$N < C_* s, \quad V \leq \mathbb{E}X^2 \leq N \mathbb{E}X = Ns < C_* s^2. \quad (\text{C.39})$$

The second inequality uses $0 \leq X \leq N$, hence $X^2 \leq NX$.

Let $V(F) = L \sqcup R$ be a bipartition of the forest and set

$$\theta = \frac{z}{1+z}, \quad \bar{q} = 1 - \theta = \frac{1}{1+z}.$$

For $C \in \{L, R\}$, let D be the opposite color and let $\mathcal{F}_C = \sigma(\xi_u : u \in C)$. Given $\mathcal{F}_C$, write $K_C = \sum_{u \in C} \xi_u$, and let a_C be the number of vertices in D having no occupied neighbor in C . The vertices counted by a_C are mutually nonadjacent, and their occupations are conditionally independent Bernoulli variables with parameter θ . Therefore

$$X \mid \mathcal{F}_C = K_C + \text{Bin}(a_C, \theta), \quad (\text{C.40})$$

$$M_C := \mathbb{E}[X \mid \mathcal{F}_C] = K_C + \theta a_C, \quad W_C := \text{Var}(X \mid \mathcal{F}_C) = \theta \bar{q} a_C. \quad (\text{C.41})$$

These are identities in the original global Gibbs law.

For each vertex v , let A_v be its addability indicator and put

$$B_v = \xi_v + A_v, \quad h_v = \xi_v - \theta B_v.$$

Thus B_v indicates that every neighbor of v is absent. From (C.40)–(C.41), the two color residuals $\varepsilon_C = X - M_C$ satisfy

$$\varepsilon_L = \sum_{v \in R} h_v, \quad \varepsilon_R = \sum_{u \in L} h_u,$$

and hence the exact pointwise identity

$$\varepsilon_L + \varepsilon_R = X - \theta \sum_v B_v = \frac{X - zA}{1+z}. \quad (\text{C.42})$$

There is one further consequence of the same single-site toggle. If $s_R = \mathbb{E} \sum_{v \in R} \xi_v$, then

$$\Pr(\xi_v = 1) = \theta \Pr(B_v = 1) \quad (v \in R),$$

so

$$\mathbb{E} W_L = \bar{q} s_R. \quad (\text{C.43})$$

The analogous formula holds with L and R interchanged.

C.6. An exponential upper bound at the mean

Lemma C.5 (probability at the mean). There are constants $V_h < \infty$, $C_h < \infty$, and $\gamma_0 > 0$, depending only on the activity cap, such that every canonical first-recovery state with $0 < z < 27$ and $V \geq V_h$ satisfies

$$\pi(0) = \Pr(X = s) \leq C_h e^{-\gamma_0 s}. \quad (\text{C.44})$$

One may take

$$C_h = 12C_*, \quad \gamma_0 = \frac{1}{25088C_*}. \quad (\text{C.45})$$

Proof. Take $V_h = V_{\text{act}}$, so that (C.34) holds for every state under consideration. Since $A \leq N - X$, (C.37) implies

$$N - s > s + 1,$$

and, because N and s are integers,

$$N \geq 2s + 2. \quad (\text{C.46})$$

Let

$$\delta = \Pr(A \geq 5s/6 \mid X = s).$$

Using (C.37) and then (C.38),

$$s + 1 < \mathbb{E}[A \mid X = s] \leq \frac{5s}{6} + N\delta$$

gives

$$\delta > \frac{s}{6N} > \frac{1}{6C_*}. \quad (\text{C.47})$$

On the event $\{X = s, A \geq 5s/6\}$, (C.34) and (C.42) give

$$\varepsilon_L + \varepsilon_R = \frac{s - zA}{1 + z} < -\frac{s}{112}. \quad (\text{C.48})$$

Thus at least one color has $\varepsilon_C < -s/224$, equivalently $M_C - s > s/224$. By (C.47) and pigeonholing the two colors, there is one fixed $C \in \{L, R\}$ such that

$$\Pr(M_C - s > s/224 \mid X = s) > \frac{1}{12C_*}. \quad (\text{C.49})$$

For an $\mathcal{F}_C$ -environment in the event in (C.49), (C.40) and the lower-tail form of Hoeffding's inequality [Hoe63] imply

$$\begin{aligned} \Pr(X = s \mid \mathcal{F}_C) &\leq \exp \left[-\frac{2(s/224)^2}{a_C} \right] \\ &\leq \exp \left[-\frac{s}{25088C_*} \right], \end{aligned} \quad (\text{C.50})$$

because $a_C \leq N < C_*s$. If $a_C = 0$, the left side is zero on this event, so the same bound holds by convention.

Let $\mathcal{E}_C = \{M_C - s > s/224\}$. Bayes' formula and (C.50) give

$$\begin{aligned}\Pr(\mathcal{E}_C \mid X = s) &= \frac{\mathbb{E}[\mathbf{1}_{\mathcal{E}_C} \Pr(X = s \mid \mathcal{F}_C)]}{\pi(0)} \\ &\leq \frac{e^{-s/(25088C_*)}}{\pi(0)}.\end{aligned}$$

Combining this with (C.49) proves (C.44)–(C.45). $\square$

C.7. A conditional-binomial lower bound and the variance lower bound

The lower estimate complementary to the preceding Hoeffding bound is the following finite, uniform local bound.

Lemma C.6 (uniform binomial mass lower bound). There are constants $c_b, C_b > 0$ such that the following holds. Let a be a nonnegative integer, let $K, s \in \mathbb{Z}$, and let

$$\frac{3}{5} \leq \theta \leq \frac{27}{28}, \quad Y = K + \text{Bin}(a, \theta), \quad M = \mathbb{E}Y, \quad W = \text{Var } Y.$$

If $H \geq 1$, $W \geq 4H$, and

$$(M - s)^2 \leq HW,$$

then

$$\Pr(Y = s) \geq c_b W^{-1/2} e^{-C_b H}. \quad (\text{C.51})$$

Proof. Since $W \geq 4$, one has $a > 0$. Put $k = s - K$, $r = k/a$, and $t = k - a\theta = s - M$. The assumptions imply

$$|t| \leq \sqrt{HW} \leq W/2 = a\theta(1 - \theta)/2.$$

Consequently

$$\frac{12}{25} \leq r \leq \frac{1539}{1568} < 1.$$

Indeed, $|r - \theta| \leq \theta(1 - \theta)/2$, and the two extrema follow at the endpoints of $[3/5, 27/28]$. In particular $1 \leq k \leq a - 1$. For every integer $n \geq 1$, the elementary Stirling inequalities

$$\sqrt{2\pi} n^{n+1/2} e^{-n} \leq n! \leq e n^{n+1/2} e^{-n}$$

give, with the universal constant $c_0 = \sqrt{2\pi}/e^2$,

$$\Pr(\text{Bin}(a, \theta) = k) \geq \frac{c_0}{\sqrt{ar(1-r)}} \exp\{-aD(r\|\theta)\}, \quad (\text{C.52})$$

where

$$D(r\|\theta) = r \log \frac{r}{\theta} + (1-r) \log \frac{1-r}{1-\theta}$$

is the Bernoulli relative entropy. Both r and θ , and therefore the segment joining them, lie in the fixed interval $J = [12/25, 1539/1568]$. Since

$$\frac{\partial^2}{\partial x^2} D(x\|\theta) = \frac{1}{x(1-x)},$$

Taylor's theorem, uniformly on J , gives

$$aD(r\|\theta) \leq C \frac{t^2}{W} \leq CH.$$

Here we used $r - \theta = t/a$ and $a^{-1} = \theta(1 - \theta)/W$. On the displayed compact intervals the ratio $r(1 - r)/[\theta(1 - \theta)]$ is bounded above and below by positive absolute constants. Thus $ar(1 - r)$ is uniformly comparable to W . Substitution in (C.52) proves (C.51). $\square$

We can now complete the second main result.

Theorem C.7 (size and variance at first recovery). There are positive constants c_V, C_V, c_N, C_N and $V_{\text{sc}} < \infty$ such that every canonical first-recovery state with $0 < z < 27$ and $V \geq V_{\text{sc}}$ satisfies

$$c_V s^2 \leq V \leq C_V s^2, \quad c_N s \leq N \leq C_N s. \quad (\text{C.53})$$

In particular,

$$V = \Theta(s^2), \quad N = \Theta(s), \quad s, N = \Theta(\sqrt{V}) \quad (\text{C.54})$$

uniformly over all canonical first-recovery states satisfying $0 < z < 27$ and $V \geq V_{\text{sc}}$.

Proof. The upper bounds have already been proved: (C.38)–(C.39) give

$$N < C_* s, \quad V < C_* s^2. \quad (\text{C.55})$$

It remains to prove the quadratic lower bound for V .

Orient the bipartition so that the unrevealed color R has mean occupation $s_R \geq s/2$, and reveal L . By (C.34), $\bar{q} = 1/(1 + z) > 1/28$, so (C.43) gives

$$\mathbb{E}W_L > \frac{s}{56}. \quad (\text{C.56})$$

Also $0 \leq W_L \leq a_L/4 \leq N/4 < C_* s/4$. Therefore

$$\Pr(W_L \geq s/112) \geq \delta_* := \frac{1}{28C_*}. \quad (\text{C.57})$$

Indeed, if the probability on the left is q , then

$$\mathbb{E}W_L \leq \frac{s}{112} + \frac{C_* s}{4} q,$$

and (C.56) yields (C.57).

Let c_b, C_b be supplied by Lemma C.6 and define

$$\gamma = \min \left\{ \frac{\gamma_0}{2C_b}, \frac{1}{448} \right\}, \quad H = \gamma s. \quad (\text{C.58})$$

By (C.55), $s \rightarrow \infty$ uniformly as $V \rightarrow \infty$. Choose $V_H \geq V_h$ so that $V \geq V_H$ implies $H \geq 1$. For either color $D \in \{L, R\}$, put

$$\mathcal{B}_D = \{W_D \geq 4H, (M_D - s)^2 \leq HW_D\}.$$

Applying Lemma C.6 conditionally in (C.40), and using $W_D \leq C_*s/4$, gives

$$\begin{aligned} \pi(0) &= \mathbb{E} \Pr(X = s \mid \mathcal{F}_D) \\ &\geq c_b e^{-C_b H} \mathbb{E}[\mathbf{1}_{\mathcal{B}_D} W_D^{-1/2}] \\ &\geq \frac{2c_b}{\sqrt{C_*s}} e^{-C_b H} \Pr(\mathcal{B}_D). \end{aligned} \quad (\text{C.59})$$

The exponential center bound (C.44), together with (C.58), now yields

$$\Pr(\mathcal{B}_D) \leq C\sqrt{s} e^{-(\gamma_0 - C_b\gamma)s} = o(1). \quad (\text{C.60})$$

Because $\gamma_0 - C_b\gamma \geq \gamma_0/2$, the bound in (C.60) tends to zero uniformly as $V \rightarrow \infty$. Fix $V_{\text{fb}} \geq V_H$ such that its right-hand side is at most $\delta_*/2$ whenever $V \geq V_{\text{fb}}$. Because $\gamma \leq 1/448$, one has $s/112 \geq 4H$. Combining (C.57) and (C.60), every state with $V \geq V_{\text{fb}}$ satisfies, with probability at least $\delta_*/2$,

$$W_L \geq \frac{s}{112}, \quad (M_L - s)^2 > HW_L \geq \frac{\gamma s^2}{112}. \quad (\text{C.61})$$

Since $\mathbb{E}M_L = \mathbb{E}X = s$, the law of total variance gives

$$\begin{aligned} V &= \mathbb{E}W_L + \text{Var}(M_L) \geq \text{Var}(M_L) \\ &= \mathbb{E}(M_L - s)^2 \geq \frac{\delta_*\gamma}{224} s^2. \end{aligned} \quad (\text{C.62})$$

Thus one may take

$$c_V = \frac{\delta_*\gamma}{224}, \quad C_V = C_*.$$

Finally, (C.46) and (C.55) in particular give

$$s \leq N < C_*s,$$

so $c_N = 1$ and $C_N = C_*$ are valid. Taking $V_{\text{sc}} = V_{\text{fb}}$ completes the proof of (C.53). The two estimates in (C.53) imply (C.54). $\square$

Appendix D. A vertex carrying a fixed variance fraction

This appendix proves the fixed-variance-fraction theorem used in Theorem 4.1. The theorem holds on the full componentwise rooted forest, for every choice of one root in each component. All conditional occupation probabilities, boundary-conditioned laws, and variances below belong to the original forest at its canonical activity. In particular, no component or descendant subtree is assigned a canonical activity or a first-recovery hypothesis.

The proof is by contradiction. Lemma D.4 excludes a diffuse exposed antichain carrying a fixed proportion of the variance. Lemma D.5 therefore reduces the variance to a finite-branch rooted subforest. Lemmas D.6 and D.7 verify the Lindeberg and predictable-quadratic-variation hypotheses there, and Proposition D.8 combines the resulting martingale central limit theorem with the full-domain Fourier envelope of Theorem A.1. The limiting positive central curvature contradicts first recovery and yields Theorem D.1.

D.1. Boundary-conditioned laws and recursive variance mass

Let

$$I_F(t) = \sum_k i_k(F) t^k$$

be the independence polynomial of a finite forest F . Suppose its coefficient sequence falls and later rises, and define

$$m = \min\{k : i_k > i_{k+1}\}, \quad s = \min\{k > m : i_k < i_{k+1}\}.$$

Thus s is the first-recovery index and $i_s^2 < i_{s-1}i_{s+1}$. The canonical activity is the unique $z > 0$ for which the hard-core law

$$\Pr_z(\xi) = \frac{z^{\sum_{u \in F} \xi_u}}{I_F(z)} \mathbf{1}_{\{\{u: \xi_u=1\} \text{ is independent}\}}$$

satisfies

$$X = \sum_{u \in F} \xi_u, \quad \mathbb{E}_z X = s, \quad V = \text{Var}_z X. \quad (\text{D.1})$$

For $j \in \mathbb{Z}$, put

$$\pi(j) = \Pr_z(X = s + j).$$

The activity factors cancel from the adjacent log-concavity difference, and hence every canonical first-recovery state satisfies

$$\pi(0)^2 - \pi(-1)\pi(1) < 0. \quad (\text{D.2})$$

Fix an arbitrary root in each component of F , and let $\mathcal{R}(F)$ be the set of component roots. For a vertex u , let T_u be its descendant subtree and let $\text{ch}(u)$ be its set of children. Every component root has a virtual parent fixed to be absent. Conditional on the parent of u being absent, write

$$p_u = \Pr_z(\xi_u = 1 \mid \xi_{\text{par}(u)} = 0), \quad q_u = 1 - p_u. \quad (\text{D.3})$$

If $R_u = p_u/q_u$ denotes the occupation odds, the rooted-tree recursion is

$$R_u = z \prod_{v \in \text{ch}(u)} q_v. \quad (\text{D.4})$$

Throughout the appendix $0 < z < 27$. Consequently,

$$0 < p_u < \bar{p}_{27} := \frac{27}{28}, \quad q_u > \frac{1}{28}. \quad (\text{D.5})$$

Let $P_u, Q_u, R_u^{\text{occ}}$ denote the laws of the occupation count in T_u when the root u is, respectively, free, forced absent, and forced occupied. Superscripts P, Q, R on means and variances refer to these three boundary-conditioned laws. Define the conditional mean difference

$$\Delta_u = \mu_u^R - \mu_u^Q.$$

Conditioning first at u and then at its children gives

$$\Delta_u = 1 - \sum_{v \in \text{ch}(u)} p_v \Delta_v. \quad (\text{D.6})$$

The parent-state probabilities are

$$a_u = \Pr_z(\xi_{\text{par}(u)} = 0), \quad b_u = 1 - a_u, \quad (\text{D.7})$$

with $a_r = 1$ and $b_r = 0$ for every $r \in \mathcal{R}(F)$. If v is a child of u , then

$$a_v = a_u q_u + b_u = 1 - a_u p_u > \frac{1}{28}, \quad b_v = a_u p_u. \quad (\text{D.8})$$

Define the **vertex variance contribution** and **subtree variance mass** by

$$g(u) = a_u p_u q_u \Delta_u^2, \quad E(u) = a_u v_u^P + b_u v_u^Q. \quad (\text{D.9})$$

At a component root r , $E(r) = v_r^P$, and component independence gives $\sum_{r \in \mathcal{R}(F)} E(r) = V$.

Recursive variance decomposition. For every vertex u ,

$$v_u^P = q_u v_u^Q + p_u v_u^R + p_u q_u \Delta_u^2, \quad v_u^Q = \sum_{v \in \text{ch}(u)} v_v^P, \quad v_u^R = \sum_{v \in \text{ch}(u)} v_v^Q. \quad (\text{D.10})$$

Moreover,

$$v_u^P \leq 28E(u), \quad v_u^Q \leq 784E(u), \quad (\text{D.11})$$

and the exact recursive decomposition is

$$E(u) = g(u) + \sum_{v \in \text{ch}(u)} E(v). \quad (\text{D.12})$$

Consequently,

$$E(u) = \sum_{v \succeq u} g(v), \quad V = \sum_{u \in F} g(u). \quad (\text{D.13})$$

Here $v \succeq u$ means that v is a descendant of u , with $u \succeq u$.

If A is an antichain, then its descendant subtrees are disjoint and

$$\sum_{u \in A} E(u) \leq V. \quad (\text{D.14})$$

Proof. The first identity in (D.10) is the conditional-variance formula for the mixture $P_u = q_u Q_u + p_u R_u^{\text{occ}}$. Under Q_u , the child subtrees have their independent P_v laws; under R_u^{occ} , they have their independent Q_v laws. This proves all of (D.10).

The mixture identity gives $v_u^P \geq q_u v_u^Q$. Also $a_u \geq 1/28$, including every component root, by (D.8). Therefore

$$v_u^P \leq a_u^{-1} E(u) \leq 28E(u), \quad v_u^Q \leq q_u^{-1} v_u^P \leq 784E(u),$$

which is (D.11). Substituting (D.10) and (D.8) into (D.9) gives

$$\begin{aligned} \sum_{v \in \text{ch}(u)} E(v) &= \sum_v [(a_u q_u + b_u) v_v^P + a_u p_u v_v^Q] \\ &= a_u (v_u^P - p_u q_u \Delta_u^2) + b_u v_u^Q = E(u) - g(u), \end{aligned}$$

proving (D.12). Iteration to the leaves proves (D.13), and summing (D.13) over disjoint descendant subtrees proves (D.14). $\square$

We can now state the full-forest theorem.

Theorem D.1 (a vertex carrying a fixed variance fraction). There are constants $\kappa_{27} > 0$ and $V_{27} < \infty$ such that every canonical first-recovery state with $0 < z < 27$ and $V \geq V_{27}$ has the following property: for every choice of one root in each component of F , some vertex $u \in F$ satisfies

$$g(u) \geq \kappa_{27} V. \quad (\text{D.15})$$

The factor a_u in $g(u)$ is always the one induced by the selected componentwise rooting of the original state.

D.2. Martingale decomposition of the occupation count

For every vertex u , define

$$\eta_u = \xi_u - p_u(1 - \xi_{\text{par}(u)}), \quad (\text{D.16})$$

using the virtual absent parent at every component root.

Lemma D.2 (martingale projection). Let $S \subseteq F$ be ancestor-closed in every rooted component. Then

$$X - s = \sum_{u \in F} \Delta_u \eta_u, \quad Z_S := \mathbb{E}[X - s \mid \xi_S] = \sum_{u \in S} \Delta_u \eta_u. \quad (\text{D.17})$$

Moreover,

$$\text{Var}(Z_S) = \sum_{u \in S} g(u), \quad \mathbb{E}[(X - s - Z_S)^2] = V - \sum_{u \in S} g(u). \quad (\text{D.18})$$

Proof. List the vertices in a parent-before-child order and let $\mathcal{F}_k$ be the sigma-field generated by the first k occupation variables, after concatenating arbitrary component orders. If u is the k -th vertex, the spatial Markov property of the forest and independence of its components give

$$\mathbb{E}[\eta_u \mid \mathcal{F}_{k-1}] = 0, \quad \mathbb{E}\eta_u^2 = a_u p_u q_u. \quad (\text{D.19})$$

Thus the weighted variables $\Delta_u \eta_u$ are martingale differences. On expanding their sum, the coefficient of ξ_u is

$$\Delta_u + \sum_{v \in \text{ch}(u)} p_v \Delta_v = 1$$

by (D.6). The remaining constant is $-\mathbb{E}X = -s$, because the sum is centered. This proves the first identity in (D.17).

Since S is ancestor-closed, a parent-before-child ordering can list all of S first. Every omitted term is then a future martingale difference, so conditional expectation proves the second identity in (D.17). Orthogonality, (D.9), and (D.19) prove the first identity in (D.18); the second is the Pythagorean identity for conditional expectation. $\square$

D.3. Diffuse exposed antichains

If $D \subseteq F$ is ancestor-closed in every rooted component, its exposed boundary is

$$\partial^+ D = \{u \notin D : u \in \mathcal{R}(F) \text{ or } \text{par}(u) \in D\}. \quad (\text{D.20})$$

An antichain A is called an **exposed antichain** if $A = \partial^+ D$ for some set D that is ancestor-closed in every rooted component. Thus A consists of the roots of the descendant components cut off from D , together with the roots of any original components disjoint from D . Its boundary-conditioned laws and weights are always inherited from the original rooted hard-core measure. For the nonnegative quantities below, a maximum over the empty set is defined to be zero.

We will also use the full-domain Fourier envelope from Appendix A, Theorem A.1. There are constants $c_{27} > 0$ and $\alpha_{27} \in (0, 1)$ such that every finite forest at activity $0 < z < 27$ satisfies

$$|\mathbb{E}_z e^{i\theta(X - \mathbb{E}X)}| \leq \exp\{-c_{27} \min(V\theta^2, (V\theta^2)^{\alpha_{27}})\}, \quad |\theta| \leq \pi. \quad (\text{D.21})$$

Lemma D.3 (antichain factorization). Let A be an exposed antichain in a rooted forest, and reveal all occupation variables outside $\bigcup_{u \in A} T_u$. Conditionally on the revealed sigma-field, the counts in the subtrees T_u , $u \in A$, are independent. The count in T_u has its P_u law when the parent of u is absent and its Q_u law when that parent is occupied. For a component root in A , the virtual parent is absent and the subtree count is deterministically in its P_u state. In all cases the unconditional probabilities of the two alternatives are a_u and b_u .

Proof. Cut the edge immediately above every nonroot member of A ; members that are component roots already begin disjoint components. The hard-core weight then factors over the resulting descendant components once the occupations in their complement, and hence all nontrivial boundary parent states, have been fixed. No parameter changes under this conditioning. The last assertion is exactly the probabilistic definition (D.7), including $a_u = 1, b_u = 0$ at a component root. $\square$

Lemma D.4 (diffuse-antichain exclusion). Let (F_n, s_n, z_n, V_n) be a canonical first-recovery sequence with $0 < z_n < 27$, $V_n \rightarrow \infty$, and an arbitrary selected componentwise rooting for each n . For every sequence of exposed antichains A_n ,

$$\max_{u \in A_n} \frac{E_n(u)}{V_n} \rightarrow 0 \quad \implies \quad \frac{1}{V_n} \sum_{u \in A_n} E_n(u) \rightarrow 0. \quad (\text{D.22})$$

Proof. Suppose otherwise. After passing to a subsequence, there is a fixed $c > 0$ such that, with

$$L_n = \sum_{u \in A_n} E_n(u), \quad \beta_n = \max_{u \in A_n} \frac{E_n(u)}{V_n}, \quad (\text{D.23})$$

one has $L_n \geq cV_n$ and $\beta_n \rightarrow 0$. By (D.14), $L_n \leq V_n$.

Appendix A, Lemma A.3, applied with activity ceiling 27, gives a constant $K_4 < \infty$ such that every P - and Q -boundary-conditioned law satisfies

$$\mathbb{E}|X_u^C - \mu_u^C|^4 \leq K_4 v_u^C (1 + v_u^C), \quad C \in \{P, Q\}. \quad (\text{D.24})$$

Indeed, the P -law is the hard-core law on the tree T_u , and the Q -law is the hard-core law on the forest below the forced absent root. Both retain the original activity.

Reveal every occupation variable outside the disjoint union of the descendant subtrees T_u , $u \in A_n$, and denote the resulting sigma-field by $\mathcal{G}_n$. By Lemma D.3, the selected subtree counts are conditionally independent P/Q blocks. If

$$M_n = \mathbb{E}[X_n \mid \mathcal{G}_n],$$

then

$$X_n - M_n = \sum_{u \in A_n} Z_{n,u}, \quad W_n = \sum_{u \in A_n} \sigma_{n,u}^2, \quad (\text{D.25})$$

where, conditionally on $\mathcal{G}_n$, the $Z_{n,u}$ are independent and centered, and $\sigma_{n,u}^2 = \text{Var}(Z_{n,u} \mid \mathcal{G}_n)$. The probabilities of the two parent states are exactly a_u and b_u , so

$$\mathbb{E}W_n = L_n \geq cV_n, \quad \text{Var}(M_n) = V_n - L_n \leq V_n. \quad (\text{D.26})$$

Equations (D.11) and (D.14) also give, for every revealed configuration,

$$0 \leq W_n \leq C_D L_n \leq C_D V_n, \quad \max_{u \in A_n} \frac{\sigma_{n,u}^2}{V_n} \leq C_D \beta_n, \quad C_D := 784. \quad (\text{D.27})$$

For fixed $\varepsilon > 0$, define the conditional Lindeberg sum

$$\Lambda_n(\varepsilon) = \frac{1}{V_n} \sum_{u \in A_n} \mathbb{E} \left[Z_{n,u}^2; |Z_{n,u}| > \varepsilon \sqrt{V_n} \mid \mathcal{G}_n \right]. \quad (\text{D.28})$$

Fourth-moment Markov, (D.11), and (D.24) imply

$$\begin{aligned} \mathbb{E}\Lambda_n(\varepsilon) &\leq \frac{K_4}{\varepsilon^2 V_n^2} \sum_{u \in A_n} [a_u v_u^P (1 + v_u^P) + b_u v_u^Q (1 + v_u^Q)] \\ &\leq \frac{K_4}{\varepsilon^2} \left(\frac{1}{V_n} + C_D \beta_n \right) \rightarrow 0. \end{aligned} \quad (\text{D.29})$$

For the second inequality, (D.11) gives

$$a_u (v_u^P)^2 + b_u (v_u^Q)^2 \leq C_D E_n(u)^2,$$

while

$$\sum_{u \in A_n} E_n(u)^2 \leq \left(\max_{u \in A_n} E_n(u) \right) L_n \leq \beta_n V_n^2.$$

Thus every term in the aggregate estimate is controlled in the original inherited normalization. This is the required aggregate conditional Lindeberg estimate.

Let

$$\Psi_n(t \mid \mathcal{G}_n) = \mathbb{E} \left[e^{it(X_n - M_n)/\sqrt{V_n}} \mid \mathcal{G}_n \right].$$

For every fixed $t \in \mathbb{R}$,

$$\mathbb{E} \left| \Psi_n(t \mid \mathcal{G}_n) - \exp \left(-\frac{t^2 W_n}{2V_n} \right) \right| \rightarrow 0. \quad (\text{D.30})$$

To prove this, Taylor-expand each centered conditional factor:

$$\mathbb{E}(e^{itZ_{n,u}/\sqrt{V_n}} \mid \mathcal{G}_n) = 1 - \frac{t^2 \sigma_{n,u}^2}{2V_n} + r_{n,u}(t).$$

For every auxiliary $\varepsilon > 0$, the cubic remainder on $|Z_{n,u}| \leq \varepsilon \sqrt{V_n}$ and the quadratic bound on its complement give

$$\sum_{u \in A_n} |r_{n,u}(t)| \leq C_t \left(\varepsilon \frac{W_n}{V_n} + \Lambda_n(\varepsilon) \right). \quad (\text{D.31})$$

Taking expectations, using (D.27) and (D.29), then sending first $n \rightarrow \infty$ and then $\varepsilon \downarrow 0$, makes the right-hand side negligible in L^1 . In addition,

$$\sum_{u \in A_n} \left(\frac{t^2 \sigma_{n,u}^2}{2V_n} \right)^2 \leq \frac{t^4}{4} \left(\max_{u \in A_n} \frac{\sigma_{n,u}^2}{V_n} \right) \frac{W_n}{V_n} \rightarrow 0 \quad (\text{D.32})$$

uniformly in the revealed configuration. The logarithmic expansion of the product of the quadratic factors now proves (D.30).

Set $\omega = c/2$. From (D.26)–(D.27),

$$\Pr(W_n \geq \omega V_n) \geq p_c := \frac{c/2}{C_D - c/2} > 0. \quad (\text{D.33})$$

Indeed, if p is the probability on the left, then $c \leq \mathbb{E}(W_n/V_n) \leq \omega + (C_D - \omega)p$. Since $\mathbb{E}M_n = \mathbb{E}X_n$, (D.26) and Chebyshev's inequality give

$$\Pr(|M_n - \mathbb{E}X_n| > B\sqrt{V_n}) \leq B^{-2}.$$

Choose a fixed B with $B^{-2} < p_c/2$, and define the $\mathcal{G}_n$ -measurable event

$$\mathcal{A}_n = \left\{ \frac{W_n}{V_n} \in [\omega, C_D], \quad \frac{|M_n - \mathbb{E}X_n|}{\sqrt{V_n}} \leq B \right\}.$$

Then

$$\Pr(\mathcal{A}_n) \geq p_c/2. \quad (\text{D.34})$$

Apply Appendix B, Theorem B.1, and pass to a further subsequence on which

$$Y_n = \frac{X_n - s_n}{\sqrt{V_n}} \Rightarrow \mu(dx) = f(x) dx,$$

where $f \in C^2(\mathbb{R})$. Put

$$m_n = \frac{M_n - \mathbb{E}X_n}{\sqrt{V_n}}, \quad w_n = \frac{W_n}{V_n}.$$

Let ν_n be the subprobability law of (m_n, w_n) restricted to $\mathcal{A}_n$. It is supported on the fixed compact rectangle $[-B, B] \times [\omega, C_D]$. After a further extraction, $\nu_n \Rightarrow \nu$, where

$$\nu(\mathbb{R}^2) \geq p_c/2. \quad (\text{D.35})$$

Let λ_n be the subprobability law of Y_n restricted to $\mathcal{A}_n$. Since $\mathbb{E}X_n = s_n$, its characteristic function satisfies, by (D.30),

$$\begin{aligned} \hat{\lambda}_n(t) &= \mathbb{E} \left[\mathbf{1}_{\mathcal{A}_n} e^{itm_n} \Psi_n(t) \mid \mathcal{G}_n \right] \\ &= \int e^{itm - t^2 w/2} d\nu_n(m, w) + o(1). \end{aligned}$$

Weak convergence of ν_n gives convergence of their total masses because their supports lie in one compact rectangle. The measures λ_n have the same total masses, as is also seen by putting $t = 0$ in their characteristic functions. These masses converge to $\nu(\mathbb{R}^2) \geq p_c/2$. After dividing λ_n and the candidate limit by their respective masses, the usual continuity theorem for probability measures [Bil99] applies; multiplying the masses back gives $\lambda_n \Rightarrow \lambda$, where

$$\lambda(dx) = h(x) dx, \quad h(x) = \int \frac{1}{\sqrt{2\pi w}} \exp\left(-\frac{(x-m)^2}{2w}\right) d\nu(m, w). \quad (\text{D.36})$$

The lower bound $w \geq \omega$ and compactness of the mixing rectangle make h continuous. In particular,

$$h(0) \geq \frac{p_c}{2\sqrt{2\pi C_D}} \exp\left(-\frac{B^2}{2\omega}\right) > 0. \quad (\text{D.37})$$

At each finite n , the full law of Y_n dominates λ_n . For every nonnegative bounded continuous function φ ,

$$\int \varphi d\mu = \lim_n \mathbb{E}\varphi(Y_n) \geq \lim_n \int \varphi d\lambda_n = \int \varphi d\lambda.$$

Hence $\mu \geq \lambda$ as measures. It follows that $f \geq h$ almost everywhere, and continuity of both densities makes the inequality pointwise. In particular, $f(0) \geq h(0) > 0$.

It remains to use the canonical center. For all sufficiently large states, Appendix C, Lemmas C.4 and C.5, give constants $C_*, C_h, \gamma_0 > 0$, depending only on the activity ceiling, such that

$$V_n \leq C_* s_n^2, \quad \pi_n(0) \leq C_h e^{-\gamma_0 s_n}. \quad (\text{D.38})$$

Since $V_n \rightarrow \infty$, these bounds imply $s_n \rightarrow \infty$ and

$$\sqrt{V_n} \pi_n(0) \leq \sqrt{C_*} C_h s_n e^{-\gamma_0 s_n} \rightarrow 0. \quad (\text{D.39})$$

On the other hand, the weighted characteristic-function convergence in Appendix B, Theorem B.1, and lattice inversion at the exact integer mean s_n give

$$\sqrt{V_n} \pi_n(0) \rightarrow f(0). \quad (\text{D.40})$$

Thus $f(0) = 0$, contradicting (D.37). This proves (D.22). $\square$

D.4. Finite-branch approximation of the variance mass

Consider any canonical first-recovery sequence with $0 < z_n < 27$, $V_n \rightarrow \infty$, and any selected componentwise rooting of each forest. On the full rooted forest define

$$S_n(\alpha) = \{u \in F_n : E_n(u) \geq \alpha V_n\}, \quad R_n(\alpha) = \sum_{u \notin S_n(\alpha)} g_n(u). \quad (\text{D.41})$$

For $0 < \alpha \leq 1$, the set $S_n(\alpha)$ is ancestor-closed inside each component because (D.12) makes E_n nonincreasing along descendant paths. A component root belongs to $S_n(\alpha)$ exactly

when its root variance mass is at least αV_n ; entire components of smaller mass may be omitted.

Lemma D.5 (finite-branch approximation). Let (F_n, s_n, z_n, V_n) be a canonical first-recovery sequence with $0 < z_n < 27$ and $V_n \rightarrow \infty$, choose one arbitrary root in every component of every F_n , and define $S_n(\alpha)$ and $R_n(\alpha)$ by (D.41). For every $\varepsilon > 0$ there are a fixed $\alpha \in (0, 1]$ and n_0 such that

$$R_n(\alpha) \leq \varepsilon V_n \quad (n \geq n_0). \quad (\text{D.42})$$

For every n , the rooted forest $S_n(\alpha)$ has at most $1/\alpha$ rooted leaves in total and is the union of at most $\lceil 1/\alpha \rceil$ componentwise root-to-leaf paths.

Proof. Let $O_n(\alpha)$ be the roots of the descendant components of $F_n \setminus S_n(\alpha)$. Thus $O_n(\alpha)$ contains every first vertex below the retained set and also the root of each original component disjoint from the retained set. This is an exposed antichain, every $w \in O_n(\alpha)$ satisfies $E_n(w) < \alpha V_n$, and (D.13) gives the exact identity

$$R_n(\alpha) = \sum_{w \in O_n(\alpha)} E_n(w). \quad (\text{D.43})$$

Indeed, every descendant of such a w also lies outside $S_n(\alpha)$, because the subtree variance mass decreases along descendant paths. The descendant subtrees rooted at $O_n(\alpha)$ therefore partition the complement of $S_n(\alpha)$.

If (D.42) failed, some $\varepsilon_0 > 0$ would permit a strictly increasing sequence n_j such that

$$R_{n_j}(1/j) > \varepsilon_0 V_{n_j}.$$

For $A_j = O_{n_j}(1/j)$, one would then have

$$\max_{w \in A_j} \frac{E_{n_j}(w)}{V_{n_j}} < \frac{1}{j}, \quad \frac{1}{V_{n_j}} \sum_{w \in A_j} E_{n_j}(w) > \varepsilon_0, \quad (\text{D.44})$$

contrary to Lemma D.4. This proves (D.42).

If $\mathcal{L}_n(\alpha)$ is the set of rooted leaves of $S_n(\alpha)$, then it is an antichain and each of its members has variance mass at least αV_n . Therefore (D.14) gives

$$|\mathcal{L}_n(\alpha)| \alpha V_n \leq \sum_{\ell \in \mathcal{L}_n(\alpha)} E_n(\ell) \leq V_n. \quad (\text{D.45})$$

Every finite rooted forest is the union of its componentwise root-to-leaf paths, proving the last assertion. $\square$

The identity (D.43) is why the subtree variance mass is used only at the exposed component roots. The interior mass $R_n(\alpha)$ is a sum of the nonnegative vertex variance contributions, each counted exactly once.

D.5. Large conditional mean differences

The next estimate converts the Feller condition $\max_u g(u)/V \rightarrow 0$ into a Lindeberg condition for the martingale array.

Lemma D.6 (large-displacement contribution bound). Let F be a finite rooted forest at activity $0 < z < 27$, with V , g , and Δ defined by (D.1)–(D.13). Let

$$g_{\max} = \max_{u \in F} g(u), \quad G_S(b) = \sum_{\substack{u \in S \\ |\Delta_u| > b}} g(u), \quad b > 1. \quad (\text{D.46})$$

If $x = 784g_{\max}/b^2 < 1$, then, for every $S \subseteq F$,

$$G_S(b) \leq \left(\frac{b}{b-1}\right)^2 28x \log\left(\frac{27}{x}\right) V. \quad (\text{D.47})$$

Consequently, let (F_n, z_n, V_n) be any sequence of such rooted forest states with $0 < z_n < 27$ and $V_n \rightarrow \infty$, let $g_{\max,n} = \max_{u \in F_n} g_n(u)$, and suppose $g_{\max,n}/V_n \rightarrow 0$. Then for every $\varepsilon > 0$ and every sequence of componentwise ancestor-closed sets $S_n \subseteq F_n$,

$$\frac{1}{V_n} \sum_{u \in S_n} \mathbb{E} \left[(\Delta_{n,u} \eta_{n,u})^2 \mathbf{1}_{\{|\Delta_{n,u} \eta_{n,u}| > \varepsilon \sqrt{V_n}\}} \right] \rightarrow 0. \quad (\text{D.48})$$

Proof. If $|\Delta_u| > b$, then (D.5), (D.8), and (D.9) give

$$p_u = \frac{g(u)}{a_u q_u \Delta_u^2} < \frac{784g(u)}{b^2} \leq x. \quad (\text{D.49})$$

For arbitrary real numbers y_v indexed by the children of u , weighted Cauchy–Schwarz gives

$$\left(\sum_v p_v y_v \right)^2 \leq \left(\sum_v \frac{p_v}{q_v} \right) \left(\sum_v p_v q_v y_v^2 \right). \quad (\text{D.50})$$

All children v of u have $a_v = 1 - a_u p_u$, and $a_u p_u q_u / a_v \leq p_u$. Multiplying (D.50) by $a_u p_u q_u$ therefore gives

$$a_u p_u q_u \left(\sum_v p_v y_v \right)^2 \leq p_u \left(\sum_v \frac{p_v}{q_v} \right) \sum_v a_v p_v q_v y_v^2. \quad (\text{D.51})$$

Equations (D.4)–(D.5) imply

$$\sum_v \frac{p_v}{q_v} \leq 28 \sum_v p_v \leq 28 \sum_v (-\log q_v) = 28 \log \left(\frac{z q_u}{p_u} \right) \leq 28 \log \left(\frac{27}{p_u} \right). \quad (\text{D.52})$$

For a counted u , (D.6) gives

$$\left| \sum_v p_v \Delta_v \right| = |1 - \Delta_u| \geq \left(1 - \frac{1}{b} \right) |\Delta_u|. \quad (\text{D.53})$$

Apply (D.51) with $y_v = \Delta_v$. Since $p \mapsto p \log(27/p)$ is increasing on $(0, 1)$, equations (D.49)–(D.53) yield

$$g(u) \leq \left(\frac{b}{b-1}\right)^2 28x \log\left(\frac{27}{x}\right) \sum_{v \in \text{ch}(u)} g(v). \quad (\text{D.54})$$

Sum over the counted parents in S . Each child contribution occurs on the right at most once, and the double sum is at most V by (D.13), proving (D.47).

Since $|\eta_u| \leq 1$, the event in (D.48) implies $|\Delta_u| > \varepsilon \sqrt{V_n}$. Take $b = \varepsilon \sqrt{V_n}$ and $x = 784g_{\max,n}/(\varepsilon^2 V_n)$. Then $x \rightarrow 0$ and $x \log(27/x) \rightarrow 0$, while the left side of (D.48) is bounded by $G_{S_n}(b)/V_n$. Equation (D.47) completes the proof. $\square$

D.6. Predictable quadratic variation on a finite-branch subtree

Standing hypotheses for Section D.6. Let F be a finite rooted forest under the hard-core law at activity $0 < z < 27$. Let $S \subseteq F$ be a finite componentwise ancestor-closed rooted subforest with L rooted leaves in total, and let $\mathcal{R}(S)$ denote its component roots. Retain the original quantities $p_u, \eta_u, \Delta_u, g(u)$ for $u \in S$. Integrating out every omitted descendant component gives an inhomogeneous hard-core law on S , with effective activities $0 < \lambda_x \leq z < 27$. More precisely, if $\mathcal{O}_x$ is the set of children of x whose descendant subtrees are omitted from S , and $\text{ch}_S(x)$ denotes its retained children, then

$$\lambda_x = z \prod_{c \in \mathcal{O}_x} \frac{Q_c(z)}{P_c(z)} = z \prod_{c \in \mathcal{O}_x} q_c \leq z.$$

Indeed, an omitted child subtree contributes $P_c(z)$ when x is absent and $Q_c(z)$ when x is occupied. For either orientation $y \rightarrow x$ of an edge, the directed-edge occupation odds in this marginal law satisfy

$$R_{y \rightarrow x} = \lambda_y \prod_{v \in N_S(y) \setminus \{x\}} (1 + R_{v \rightarrow y})^{-1} < 27. \quad (\text{D.55})$$

Therefore

$$\Pr(\xi_y = 1 \mid \xi_x = 0) < \bar{p}_{27}, \quad \Pr(\xi_y = 1 \mid \xi_x = 1) = 0 \quad (\text{D.56})$$

in either direction along every edge of S .

The marginalization also preserves the original conditional occupation probability at a retained vertex. Conditional on its parent being absent, the odds at u in the marginal law are

$$\lambda_u \prod_{v \in \text{ch}_S(u)} q_v = z \prod_{v \in \text{ch}_F(u)} q_v = R_u = \frac{p_u}{q_u}.$$

Thus

$$\Pr(\xi_u = 1 \mid \xi_{\text{par}(u)} = 0) = p_u.$$

For a parent-before-child filtration on S , all unrevealed descendants have already been integrated into these odds, while the only previously revealed neighbor of u is its parent. Let $\mathcal{F}_{u-}^S$ denote the sigma-field generated by the retained vertices preceding u . Consequently

$$\mathbb{E}(\xi_u \mid \mathcal{F}_{u-}^S) = p_u(1 - \xi_{\text{par}(u)}).$$

It follows directly that the original variable $\eta_u = \xi_u - p_u(1 - \xi_{\text{par}(u)})$ remains a martingale difference and

$$\mathbb{E}(\eta_u^2 \mid \mathcal{F}_{u-}^S) = p_u q_u(1 - \xi_{\text{par}(u)}),$$

with the virtual absent parent convention at every retained component root. Thus the summands in (D.57) are the predictable conditional variances of the projected martingale from Lemma D.2.

Lemma D.7 (quadratic-variation concentration). Under the standing hypotheses above, enumerate S in parent-before-child order and let $\mathcal{F}_k^S$ be the sigma-field generated by its first k occupation variables. The predictable quadratic variation of the projected martingale is

$$Q_S = \sum_{k=1}^{|S|} \mathbb{E}[(\Delta_{u_k} \eta_{u_k})^2 \mid \mathcal{F}_{k-1}^S]. \quad (\text{D.57})$$

If $g_{\max} = \max_{u \in S} g(u)$, then

$$\mathbb{E}Q_S = \sum_{u \in S} g(u), \quad \text{Var}(Q_S) \leq C_{27} L^2 g_{\max} V, \quad C_{27} = 10\,976. \quad (\text{D.58})$$

Proof. If x and y lie in different components, their occupation variables are independent and their covariance is zero. Suppose they lie in the same component, and let $x = v_0, v_1, \dots, v_d = y$ be the unique path in S . For an oriented edge $v_j v_{j+1}$, the binary conditional expectation has the affine form

$$\mathbb{E}(\xi_{v_{j+1}} \mid \xi_{v_j}) = \alpha_j + \beta_j \xi_{v_j}, \quad \beta_j = \Pr(\xi_{v_{j+1}} = 1 \mid \xi_{v_j} = 1) - \Pr(\xi_{v_{j+1}} = 1 \mid \xi_{v_j} = 0).$$

Equation (D.56) gives $|\beta_j| < \bar{p}_{27}$. For $t \in \{0, 1\}$, write

$$m_j(t) = \mathbb{E}(\xi_{v_j} \mid \xi_x = t) = A_j + B_j t, \quad A_0 = 0, \quad B_0 = 1.$$

The forest Markov property and the tower rule give the two-line recursion

$$m_{j+1}(t) = \alpha_j + \beta_j m_j(t) = (\alpha_j + \beta_j A_j) + \beta_j B_j t.$$

Thus $B_{j+1} = \beta_j B_j$, and induction from $B_0 = 1$ gives $B_d = \prod_{j=0}^{d-1} \beta_j$. In particular, the slope of $\mathbb{E}(\xi_y \mid \xi_x)$ as an affine function of ξ_x is this product. Hence

$$\text{Cov}(\xi_x, \xi_y) = \text{Cov}(\xi_x, \mathbb{E}[\xi_y \mid \xi_x]) = \text{Var}(\xi_x) \prod_{j=0}^{d-1} \beta_j.$$

Since $\text{Var}(\xi_x) \leq 1/4$, this proves

$$|\text{Cov}(\xi_x, \xi_y)| \leq \frac{1}{4} \bar{p}_{27}^{\text{dist}(x,y)}. \quad (\text{D.59})$$

A rooted forest with L rooted leaves has at most $2L$ vertices of degree at most one in the underlying unrooted forest; these are what we call **unrooted leaves** here. Indeed, a nontrivial rooted component with l rooted leaves has at most $l + 1$ unrooted leaves, while an isolated rooted component has one; summing over at most L components gives the claim. For a fixed $x \in S$, the branches beginning at vertices at a common positive distance k from x are disjoint, and each contains an unrooted leaf. Hence

$$|\{y \in S : \text{dist}(x, y) = k\}| \leq 2L \quad (k \geq 0). \quad (\text{D.60})$$

Put

$$c_u = p_u q_u \Delta_u^2 = \frac{g(u)}{a_u} \leq 28g(u). \quad (\text{D.61})$$

For a nonroot $u = u_k$, its summand in (D.57) is $c_u(1 - \xi_{\text{par}(u)})$; every retained component-root summand is deterministic. Grouping by parents gives

$$Q_S = \sum_{r \in \mathcal{R}(S)} c_r + \sum_{x \in S} w_x (1 - \xi_x), \quad w_x = \sum_{\substack{u \in S \\ \text{par}(u) = x}} c_u. \quad (\text{D.62})$$

Every child branch in S contains a rooted leaf, so each vertex has at most L children in S . Thus

$$\max_x w_x \leq 28Lg_{\max}, \quad \sum_x w_x \leq 28V. \quad (\text{D.63})$$

Using (D.59)–(D.60) and $(1 - \bar{p}_{27})^{-1} = 28$, we obtain

$$\begin{aligned} \text{Var}(Q_S) &\leq \frac{1}{4} \sum_x w_x \sum_y w_y \bar{p}_{27}^{\text{dist}(x,y)} \\ &\leq \frac{1}{4} (28V)(28Lg_{\max})(2L) \sum_{k \geq 0} \bar{p}_{27}^k \leq 10\,976 L^2 g_{\max} V. \end{aligned} \quad (\text{D.64})$$

The last numerical constant is $\frac{1}{4} \cdot 28^3 \cdot 2 = 10\,976$.

Finally, $\mathbb{E}(1 - \xi_{\text{par}(u)}) = a_u$, so (D.61)–(D.62) give $\mathbb{E}Q_S = \sum_{u \in S} a_u c_u = \sum_{u \in S} g(u)$. This proves (D.58). $\square$

D.7. Failure of the Feller condition

Proposition D.8 (failure of the Feller condition). There is no canonical first-recovery sequence with $0 < z_n < 27$, $V_n \rightarrow \infty$, and a selected componentwise rooting of each forest for which

$$g_{\max,n} := \max_{u \in F_n} g_n(u) = o(V_n). \quad (\text{D.65})$$

Proof. Assume (D.65). For every integer $j \geq 2$, apply Lemma D.5 with $\varepsilon = 1/j$, lower the resulting fixed threshold if necessary so that $\alpha_j \leq 1/j$, and put

$$\ell_j = \lceil 1/\alpha_j \rceil.$$

There is a strictly increasing sequence $N_j \geq j$ such that, for every $n \geq N_j$,

$$\frac{R_n(\alpha_j)}{V_n} \leq \frac{1}{j}, \quad \frac{\ell_j^2 g_{\max, n}}{V_n} \leq \frac{1}{j}. \quad (\text{D.66})$$

For $n \geq N_2$, let

$$j(n) = \max\{j : N_j \leq n\}, \quad S_n = S_n(\alpha_{j(n)}).$$

Then $j(n) \rightarrow \infty$. If ℓ_n is the number of rooted leaves of S_n , Lemma D.5 and (D.43) give

$$\sum_{u \in S_n} g_n(u) = V_n - R_n(\alpha_{j(n)}) = (1 - o(1))V_n, \quad \frac{\ell_n^2 g_{\max, n}}{V_n} \rightarrow 0. \quad (\text{D.67})$$

Both conclusions follow with $j = j(n)$: one has $n \geq N_{j(n)}$, while Lemma D.5 gives $\ell_n \leq \ell_{j(n)}$. Thus no estimate obtained for a fixed threshold is used before its own eventual index.

Let $Q_n = Q_{S_n}$. Lemma D.7 implies

$$\frac{\mathbb{E}Q_n}{V_n} \rightarrow 1, \quad \text{Var} \left(\frac{Q_n}{V_n} \right) \leq C_{27} \frac{\ell_n^2 g_{\max, n}}{V_n} \rightarrow 0.$$

Therefore

$$\frac{Q_n}{V_n} \rightarrow 1 \quad \text{in probability.} \quad (\text{D.68})$$

Write $S_n = \{u_{n,1}, \dots, u_{n,M_n}\}$ in parent-before-child order, let $\mathcal{F}_{n,k}$ be generated by the first k listed occupation variables, and put $D_{n,k} = \Delta_{n,u_{n,k}} \eta_{n,u_{n,k}}$. For fixed $\varepsilon > 0$, define

$$\Gamma_n(\varepsilon) = \frac{1}{V_n} \sum_{k=1}^{M_n} \mathbb{E} \left[D_{n,k}^2 \mathbf{1}_{\{|D_{n,k}| > \varepsilon \sqrt{V_n}\}} \mid \mathcal{F}_{n,k-1} \right]. \quad (\text{D.69})$$

Its expectation tends to zero by Lemma D.6 and (D.65). Markov's inequality therefore gives

$$\Gamma_n(\varepsilon) \rightarrow 0 \quad \text{in probability.} \quad (\text{D.70})$$

These are finite square-integrable martingale-difference rows normalized by $\sqrt{V_n}$. The constant-variance martingale-array central limit theorem stated after (B.60), applied with (D.68) and (D.70), yields

$$\frac{Z_{S_n}}{\sqrt{V_n}} \Rightarrow \mathcal{N}(0, 1). \quad (\text{D.71})$$

Only the ordinary weak convergence in (D.71) is taken from the martingale central limit theorem. No local limit theorem or adjacent-probability estimate is attributed to Hall and Heyde; those estimates enter below from the separate full-domain Fourier envelope.

By Lemma D.2 and (D.67),

$$\frac{1}{V_n} \mathbb{E}[(X_n - s_n - Z_{S_n})^2] = 1 - \frac{1}{V_n} \sum_{u \in S_n} g_n(u) \longrightarrow 0. \quad (\text{D.72})$$

Slutsky's theorem transfers (D.71) to the original global law:

$$\frac{X_n - s_n}{\sqrt{V_n}} \Rightarrow \mathcal{N}(0, 1). \quad (\text{D.73})$$

Weak convergence alone does not determine the adjacent log-concavity difference. For the local step, use the full-domain envelope (D.21) and set

$$H_{27}(u) = \exp \{ -c_{27} \min(u^2, |u|^{2\alpha_{27}}) \}. \quad (\text{D.74})$$

Every polynomial weight is integrable against H_{27} . Define

$$\phi_n(u) = \mathbb{E} \exp \left\{ iu \frac{X_n - s_n}{\sqrt{V_n}} \right\}, \quad I_n = [-\pi\sqrt{V_n}, \pi\sqrt{V_n}], \quad c_G = (2\pi)^{-1/2}.$$

Equations (D.73)–(D.74) give

$$\phi_n(u) \longrightarrow e^{-u^2/2}, \quad |\phi_n(u)| \leq H_{27}(u) \quad (u \in I_n). \quad (\text{D.75})$$

Extend each Fourier integrand by zero outside I_n . Lattice inversion at the exact integer center gives

$$\sqrt{V_n} \pi_n(0) = \frac{1}{2\pi} \int_{I_n} \phi_n(u) du \longrightarrow c_G. \quad (\text{D.76})$$

Together with (D.39), equation (D.76) already gives a contradiction. For completeness, the following calculation supplies a second, independent contradiction by recovering the sign of the adjacent log-concavity determinant.

Define

$$\delta_n = \pi_n(-1) + \pi_n(1) - 2\pi_n(0), \quad \tau_n = \pi_n(1) - \pi_n(-1). \quad (\text{D.77})$$

The scaled second-difference multiplier $2V_n(\cos(u/\sqrt{V_n}) - 1)$ converges to $-u^2$ and is bounded in absolute value by u^2 . Dominated convergence with $u^2 H_{27}(u)$ gives

$$\begin{aligned} V_n^{3/2} \delta_n &= \frac{1}{2\pi} \int_{I_n} \phi_n(u) 2V_n \left(\cos \frac{u}{\sqrt{V_n}} - 1 \right) du \\ &\longrightarrow -\frac{1}{2\pi} \int_{\mathbb{R}} u^2 e^{-u^2/2} du = -c_G. \end{aligned} \quad (\text{D.78})$$

Similarly, the multiplier $-2i\sqrt{V_n}\sin(u/\sqrt{V_n})$ is bounded by $2|u|$ and converges to $-2iu$. Dominated convergence and oddness give

$$\begin{aligned} V_n \tau_n &= \frac{1}{2\pi} \int_{I_n} \phi_n(u) \left(-2i\sqrt{V_n} \sin \frac{u}{\sqrt{V_n}} \right) du \\ &\rightarrow -\frac{i}{\pi} \int_{\mathbb{R}} u e^{-u^2/2} du = 0. \end{aligned} \tag{D.79}$$

The exact relations

$$\pi_n(-1) = \pi_n(0) + \frac{\delta_n - \tau_n}{2}, \quad \pi_n(1) = \pi_n(0) + \frac{\delta_n + \tau_n}{2}$$

imply

$$\pi_n(0)^2 - \pi_n(-1)\pi_n(1) = -\pi_n(0)\delta_n - \frac{\delta_n^2 - \tau_n^2}{4}. \tag{D.80}$$

By (D.76), (D.78), and (D.79),

$$V_n^2 [\pi_n(0)^2 - \pi_n(-1)\pi_n(1)] \rightarrow c_G^2 = \frac{1}{2\pi} > 0. \tag{D.81}$$

This contradicts the canonical first-recovery sign (D.2), and proves the proposition. $\square$

D.8. Uniformization over all rootings

Proof of Theorem D.1. If the asserted constants did not exist, then for every integer $j \geq 1$ one could choose a canonical first-recovery state with $0 < z_j < 27$, $V_j \geq j$, and a choice of one root in each component of F_j such that

$$\max_{u \in F_j} \frac{g_j(u)}{V_j} < \frac{1}{j}.$$

These choices would form a sequence forbidden by Proposition D.8. Hence uniform constants $\kappa_{27} > 0$ and $V_{27} < \infty$ exist. Since the contrary sequence was allowed to select arbitrary componentwise rootings in every state, the conclusion holds for every such rooting. The resulting vertex belongs to the original forest, with the weight induced by that rooting. $\square$

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